Information, Empathy, and In-Group Solidarity among Public Kindergarten Parents: A Field and Survey Experiment in a Large German City

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Abstract

Can empathy and information increase programme beneficiaries' willingness to share the costs of public redistribution? We study parents using publicly subsidised childcare in Germany, where income-dependent fees redistribute costs across income groups. We test whether brief empathy appeals and cost information strengthen solidarity among current and future beneficiaries. Beneficiaries share an interest in maintaining the public programme but differ as to whether they are net contributors or net recipients. In a pre-registered field experiment (Study 1), 13,161 survey invitations were randomised to letters emphasising empathy or efficiency. Empathy messages did not increase participation (17.3 versus 17.6%; p = .716). In a subsequent pre-registered survey experiment (Study 2; N = 2,298), respondents received generic information, an empathy story, fiscal information, or no treatment. None significantly affected preferences for fair childcare fees or donations. Brief empathy appeals and fiscal information have limited impact in strengthening redistributive solidarity among beneficiaries of public services.

Keywords: solidarity; childcare fees; field experiment; survey experiment; empathy; information treatment

Introduction

Citizens in welfare states often benefit from redistributive programmes delivering services, infrastructure, and cash transfers. Sometimes people are aware of the policies from which they benefit, such as a monthly child payment; but usually they are not because these resources are invisible to them. This is a version of what Mettler (2011) calls the 'submerged state'. Just like tax breaks, some programmes have no impact on public knowledge and attitudes because there is no badging of the transfer or a physical impact. Because of invisibity, beneficiares are unaware of collective purposes behond such state interventions and believe that their own private actions and resources are being deployed. Moreover, public funding is a complex technical field about which citizens have little knowledge, reinforcing this invisibility. But can telling service users about these benefits, addressing this information assymmetry, build solidarity or is this lack of awareness immutable?

In this paper, we study one such group of beneficiaries: German parents whose pre-school children (0-6 years) attend publicly subsidised kindergartens. It costs about €1,000 per month per child of which parents pay on average seven per cent (BLINDED internal communication with city). Publicly supported childcare is one of the places where welfare-state solidarity becomes concrete. That makes kindergarten financing a good case for studying how citizens respond to information about redistribution. We deploy arguments that a frameed informarion message may trigger empathy thus underlying support for solidaristic financing (Batson et al., 1997; van Oorschot, 2000) It can tap into wider attitudes toward welfare-state redistribution (Kumlin and Stadelmann-Steffen, 2014; Kuziemko et al., 2015). Field experiments have examined whether information and practical assistance improve access to subsidised childcare (Hermes et al., 2025; Carbuccia et al., 2026). We extend this research by examining whether among parents already using the service.

To test the claim that information and empathy appeals strengthen solidarity, we implemented a pre-registered field and a survey experiment on kindergarten parents' behaviour and attitudes in METROPOLIS, a large German city.[1] Study 1 asked whether an empathy-oriented invitation letter brings more people into the survey than an efficiency-oriented letter. The linked Study 2 asked whether respondents differ in their views or donation behaviour after receiving generic information, an empathy story, or fiscal information about kindergarten costs, and those in the control group who received none. Study 1's outcome is agreeingto take part in the survey from the inviation lettee; Study 2's are perception of fairness of the monthly fee and willingness to give to a local children's organisation.

The empathy invitation did not increase survey take-up. Within the survey, the treatments produced small and imprecise movements in the fair monthly fee proxy and donation outcome. We did not find effect heterogeneity. We therefore confirm a null result for this case: brief empathy and information interventions may be insufficient to shift expressed solidarity in this local childcare-finance context. The wider contribution is that even in a propitious context of a homogenous group, visbility does little to uncover the submerged state.

Theory and Expectations: Information, Empathy and In-group Solidarity

In this study, we test two causal pathways that may increase in-group solidarity and thereby strengthen the community of beneficiaries of public policy: (a) providing people with factual information about the extent of the redistributive policy programme and (b) providing people with an empathetic story about co-beneficiaries.

Information and belief change. A group of policy beneficiaries needs to know about the program and about how to take part in it. A large US literature on the "submerged state" examines how the state indirectly incentivizes private behavior at a cost to taxpayers. Such non-obvious state actions create the illusion of receiving no benefits among beneficiaries (Mettler, 2011). Knowing about the programme is a necessary condition for developing group consciousness and a willingness to support that group. Lü (2014) argues, in the context of Chinese citizens, that awareness of a policy creates attitudinal change, rather than awareness of the benefits.

[1] Name withheld.

Fiscal information could increase support for collective financing by making the solidarity system visible and legitimate. Hetling et al. (2008) demonstrate for American social-policy reforms that awareness of policy reform itself—that the state does something for a certain group—already improves views of welfare recipients.

Existing work on information-provision experiments suggests that factual corrections can matter, but that they often depend on prior beliefs, attention, and motivated reasoning (Haaland et al., 2023; Kuziemko et al., 2015). Once kindergarten parents know the details of the programme, this knowledge can trigger interpretive policy-feedback effects: parents may perceive themselves as a group of beneficiaries who receive substantial general support from taxpayers and develop a sense of common status in the welfare state. Malhotra and Shi (2026) find that US small business owners increased support for anti-poverty spending by 6.9 percentage points of the aid they received when reminded. Bechtel and Scheve (2017)'s experiment shows that citizens contribute more to a public good when expecting others to give more, using a sample of four countries including Germany, findings that reach back to a considerable literature on conditional cooperation Fischbacher et al. (2001)

But providing information about the extent of the programme could also have no effect or even backfire if the same information makes the taxpayer burden more salient. Important for our study, Lergetporer et al. (2018) find that German parents are less willing to support public spending on schools as they thought it was much less. Much my depend on exactly how the information is presented, as is shown in Scotto et al. (2017)'s study of support for foreign aid.

Empathy for co-beneficiaries.

Whereas information about the extent of the programme and the size of public investment follows a cognitive, rational logic, we are also interested in the emotional triggers of in-group solidarity, particularly empathy. Empathy for an individual or a group is a multidimensional phenomenon that includes perspective-taking, an increased sense of commonality, greater concern for welfare, feelings of warmth, and the motivation to help (Batson, 2022). Empathy for a group is harder to generate because people can more readily build empathy with single individuals than with groups of similar individuals (Small et al., 2007). Aarøe and Petersen (2014) show in an exoeriment on Danish and US citizens that a story about an unlucky welfare recipient shifted steotypes and hence opposition to welfare. Empathy can matter.

Preregistered Hypotheses

The preregistration states one Study 1 hypothesis and five Study 2 hypotheses. We split Study 2 H1 into H1a and H1b below because it covers two primary outcomes.

- **Study 1 H1:** The empathy invitation increases survey take-up relative to the efficiency invitation.
- **Study 2 H1a:** The empathy-story and fiscal-information treatments increase the fair monthly fee outcome relative to control.
- **Study 2 H1b:** The empathy-story and fiscal-information treatments increase donation amounts relative to control.

- **Study 2 H2:** Within-survey information effects are stronger after the empathy invitation.
- **Study 2 H3 (exploratory):** Treatment effects are moderated by ideology, with stronger effects among more left-leaning respondents.
- **Study 2 H4 (exploratory):** Treatment effects vary by education.
- **Study 2 H5 (exploratory):** The empathy story should move donation behavior more, while fiscal information should move fair-fee preferences more.

The preregistration also identifies income moderation and the prior-belief-gap analysis as exploratory analyses without hypothesis numbers; it contains no H6.

Supplementary Appendix Sections A.3 and G.1–G.3 detail the hypothesis labels, preregistration coverage, and differences between the planned and implemented procedures.

Research Design, Treatments, and Outcomes

Research Design

Setting and two-stage design

Parents of preschool children in public kindergartens are a likely political in-group. They are of similar age, face similar demands in structuring and financing everyday life, and collectively share parental challenges. Parents of preschool children are also politically similar because they share the same position in their life course, compared with the hypothetical position of having no preschool children. This parental preference trajectory is gendered. Mothers move more strongly to the left with increasing demand for state action in childcare and early childhood education (Banducci et al., 2016).

Many citizens do not know how kindergarten places are financed or how large the public subsidy is (93 percent from VAT and income tax in our city). As our data show, kindergarten parents substantially overestimate the proportion of total costs represented by their personal contribution. This mismatch between knowledge and reality is probably due to the fact that almost all educational programmes in Germany are financed by general taxes rather than privately by citizens themselves. It is thus plausible that giving these uninformed parents more information increases their sense of belonging with other beneficiary parents.

In Germany, more than 90 percent of pre-school children attend a publicly provided kindergarten that is only minimally co-financed by the parents, a system that traditionally enjoys large support (Goerres and Tepe, 2012). Thus, such parents are even more homogenous politically as co-beneficiaries of a large redistributive policy programme.

We conducted a two-stage field and survey experiment among parents of children attending kindergarten in METROPOLIS, Germany. The study was implemented through a municipal survey of families with kindergarten-age children (0-6 years). Kindergarten provision is largely publicly financed, while parents contribute fees that vary across families, making this a setting in which solidarity toward other beneficiaries of the same public service has direct financial implications. In

the first stage, 13,161 families were randomly assigned to receive one of two postal invitation letters: an efficiency-oriented appeal or an empathy-oriented appeal. This field experiment tests whether the framing of the invitation affects subsequent survey participation.

Families who followed the invitation and entered the online survey proceeded to the second stage of the study. In this second stage, the 2,298 respondents included in the survey data were independently randomized to one of four conditions: control, generic information about kindergarten provision, an empathy-based narrative, or fiscal information about kindergarten financing. This stage was designed to capture both attitudinal and behavioral expressions of solidarity. Respondents reported what they considered a fair monthly kindergarten fee and decided how much of a potential lottery prize to allocate to a local children's organization. Figure 1 summarizes the sequence from the municipal sampling frame through the invitation experiment, survey participation, the second randomization, and the outcome-specific analysis samples.

The two experimental stages therefore concern different populations and different decisions. Study 1 estimates the effect of invitation framing on participation among all randomized invitation records. Study 2 estimates the effects of the survey treatments among respondents who provided informed consent and were retained for analysis. The Study 2 treatment was randomized separately from the invitation assignment, and the primary Study 2 analysis therefore compares the four survey-treatment conditions among respondents included in Study 2. A secondary analysis links the two stages at the respondent level to examine whether the effects of the survey treatments differ depending on the invitation condition respondents had received.

Supplementary Appendix Sections A.1–A.2 provide further context on the setting and the two study populations.

Study 1: Invitation-letter field experiment

The municipal source workbook contains 15,788 records. The invitation-preparation procedure trims the family identifier, removes 475 records with blank identifiers, and retains the first record for each remaining identifier, removing 2,152 repeated rows. This yields 13,161 distinct invitation records. These are preprocessing exclusions prior to experimental assignment and therefore do not represent participant attrition. The randomization unit is the retained family identifier used to prepare the mailing rather than each child-level source row. The survey introduction identifies the University of BLINDED as the sender of the invitation letters, and the city as its cooperation partner.

The archived randomization procedure uses a fixed random seed and first selects the addressee. Where a second addressee is available, half of the records are assigned to each addressee category; otherwise the first addressee is used. Within addressee categories, records are randomly ordered and the two message conditions alternate, producing approximately equal allocation. The efficiency invitation contains 6,581 records and the empathy invitation 6,580. Both letters ask recipients about life with kindergarten children. The efficiency text emphasizes improving the quality and efficiency of public services and developing municipal institutions. The empathy text emphasizes making METROPOLIS better for families and supporting the community of parents with kindergarten children.

Survey take-up is recovered by joining the unique invitation code to the merged Wave 1 status data. Consistent with the preregistration, take-up is defined as accessing the Wave 1 survey and

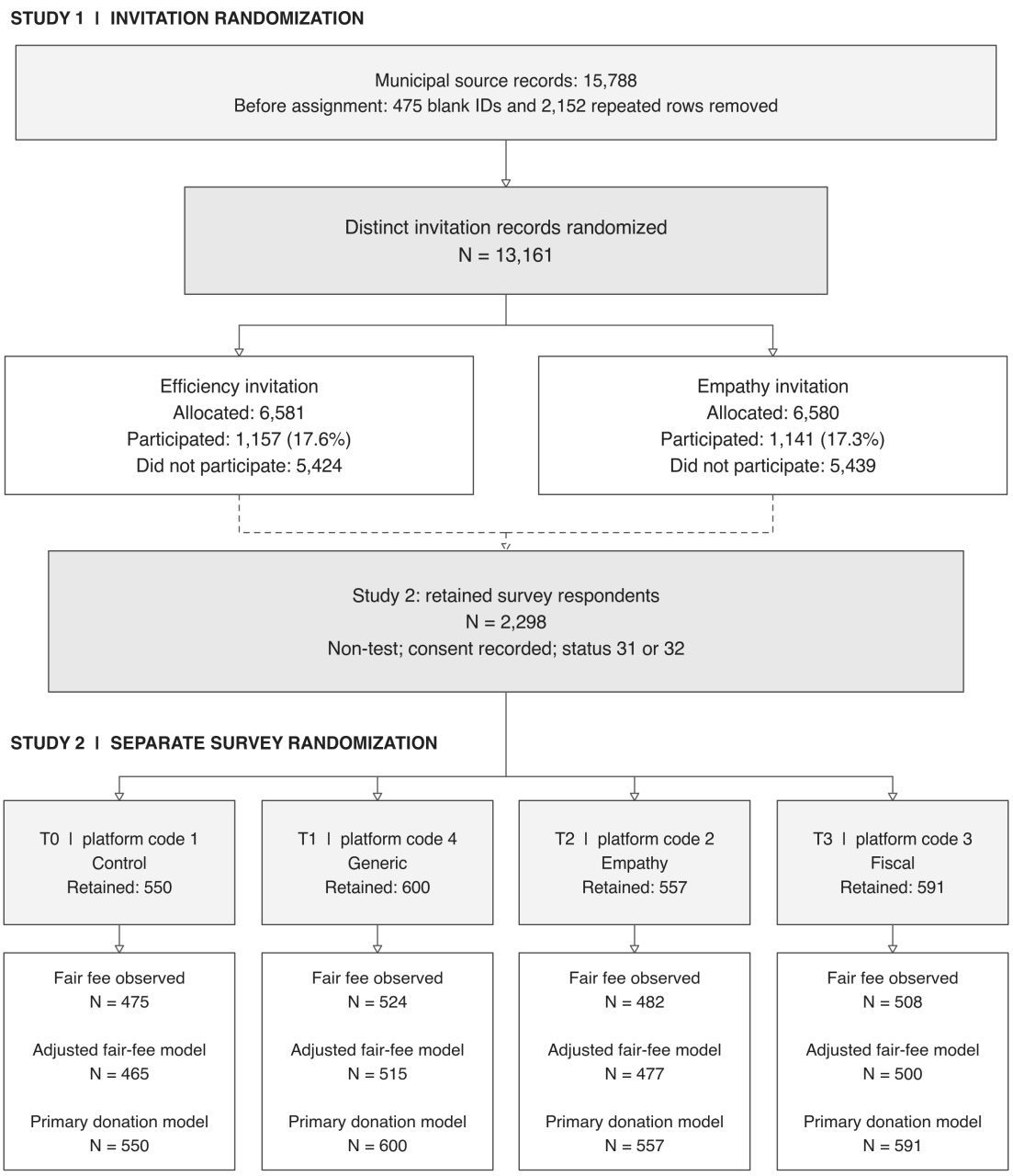


Figure 1: Participant flow through the invitation-letter field experiment and subsequent survey experiment.

providing informed consent. All Wave 1 survey respondents record affirmative informed consent. This procedure identifies 1,157 participants in the efficiency condition and 1,141 in the empathy condition.

The municipality sent the invitation letters by snail mail during the week 9–15 March 2026.

Frame preparation, addressee selection, recruitment, and participation coding are documented in Supplementary Appendix Sections B.1–B.4. Sections D.1–D.2 reproduce the invitation paragraphs and their English translations.

Study 2: Survey experiment

The dataset contains 2,298 Wave 1 survey rows. All respondents have a valid survey-treatment assignment and record affirmative informed consent.

Following the welcome page and before the consent question, respondents were independently randomized by a Unipark trigger. The trigger generated a random integer from 1 to 4 and stored it as the treatment assignment variable. No balancing across treatment arms was enforced. The survey-platform codes correspond to the preregistered treatment labels as follows: platform code 1 = T0, control ($N = 550$); platform code 4 = T1, generic information ($N = 600$); platform code 2 = T2, empathy ($N = 557$); and platform code 3 = T3, fiscal information / community of solidarity ($N = 591$). The numerical platform codes should therefore not be interpreted as corresponding sequentially to T0–T3.

The instrument contains four experimental conditions: a control condition with no additional information; a generic description of METROPOLIS kindergartens, providers, locations, and access; an empathy narrative about explicitly fictional parents Julia and Daniel and their daughter Sofia, whose kindergarten attendance is made affordable through income-related fees; and fiscal information emphasizing shared taxpayer financing and redistribution across families. The fiscal-information screen presents parental payments alongside the total cost of kindergarten provision. Baseline fee and financing-belief questions precede the experimental screens. The fairness outcome follows the intervention, while the donation decision appears later in the questionnaire.

Assignment, routing, questionnaire order, and recognition checks are detailed in Supplementary Appendix Sections C.1–C.4. Sections D.3–D.5 provide the survey screens, images, translations, and material-version comparisons; Sections M.1–M.2 supply the questionnaire transcription.

Primary outcomes

Fair monthly fee. The fair monthly fee is the monthly Euro amount that the parents deems to be fair to pay for him or her. It is a proxy that combines the respondent's current-fee category, the assessment of whether that fee is fair, and, where relevant, a conditional proposed fair fee. For respondents choosing the central "fair" response, the outcome is represented by the respondent's current-fee proxy. The no-fee category is coded as EUR 0. The ten positive categories, ranging from EUR 1–50 through EUR 451–500, are represented by EUR 25, 75, 125, 175, 225, 275, 325, 375, 425, and 475, respectively. These values correspond to the rounded bracket representatives used in the existing analysis.

For respondents judging their current fee to be unfair, the outcome is the numeric amount they state would constitute a fair fee. If the required response is missing, the outcome remains missing.

The fair-fee proxy is observed for 1,989 respondents. Because the preregistered specification additionally adjusts for the respondent's current fee, the adjusted model requires a non-missing current-fee covariate and contains 1,957 respondents. Observing the outcome and entering the adjusted analysis sample are therefore separate stages, as shown in Figure 1.

The variable map and fair-fee construction are detailed in Supplementary Appendix Sections E.1–E.2.

Donation. The donation decision asks respondents to allocate a potential EUR 100 lottery win after the survey will conclude between themselves and BLINDED-ORGANIZATION in METROPOLIS supporting children and kindergarten projects in it. Respondents choose a donation between EUR 0 and EUR 100 in EUR 10 increments and retain the remainder as a digital gift voucher if they win. The questionnaire advertises 25 prizes of EUR 100. The measure therefore represents a conditional allocation of possible lottery winnings rather than an immediate payment by every respondent.

The primary coding preserves the selected donation amount when the respondent also selects a lottery contact option. Explicit lottery opt-outs at either the allocation or contact question are coded as EUR 0, following the preregistration used by the current analysis. Other unresolved missing responses are not automatically coded as zero. In the dataset, 2,040 respondents confirm lottery participation and 258 explicitly opt out, yielding 2,298 observations for the primary donation outcome. The 2,198 usable responses to the allocation item alone therefore do not define the primary analysis sample. Conditional donation analyses and financing satisfaction are treated as secondary outcomes.

Supplementary Appendix Sections E.3–E.5 describe donation coding, secondary measures, moderators, and missing-value handling.

Linkage, preregistration, and ethics

Invitation-by-survey-treatment analyses use respondent-level linkage between the invitation randomization and the survey response data. Wave 1 survey responses were linked to the corresponding randomized invitation records using the unique invitation code. This linkage connects each retained survey respondent to the invitation condition assigned in Study 1. Outcome and covariate requirements then determine the usable sample for each model. Figure 1 distinguishes invitation records, retained survey respondents, randomized survey arms, and outcome-specific analysis samples.

Outcome-specific sample counts, data-version comparisons, and linkage diagnostics are reported in Supplementary Appendix Sections F.1–F.4. Sections H.1–H.4 detail the estimands, model specifications, multiple-comparison adjustments, and interaction interpretations.

The preregistration distinguishes the two primary survey outcomes from secondary and exploratory analyses. Participation and data use require informed consent. The survey explains that participation is voluntary and confidential, permits withdrawal, and separately requests permission for

further contact. The preregistration described Study 2 randomization as occurring among consenting respondents. In the implemented survey, however, treatment assignment was generated on the first survey page, before the consent question. The analysis nevertheless includes only respondents who provided informed consent.

The study received ethical approval from the Ethics Committee of BLINDED (approval no. 2602PWGA7734, 4 March 2026).

The study was preregistered prior to analysis; the anonymized preregistration is available on OSF.

Further ethics and consent documentation is organized in Supplementary Appendix Sections J.1–J.7. Sections K.1–K.3 and L.1–L.6 cover replication materials, source provenance, and the reporting-checklist responses.

Results

Empirical specifications

For Study 1, we report the empathy-minus-efficiency difference in survey take-up and the corresponding odds ratio from

$$\text{logit}\{\Pr(TakeUp_i = 1)\} = \alpha + \beta EmpathyLetter_i. \tag{1}$$

For Study 2, T0 (no additional information) is the reference arm. The confirmatory specifications are

$$FairFee_i = \alpha + \beta_1 T1_i + \beta_2 T2_i + \beta_3 T3_i + \gamma CurrentFee_i + \varepsilon_i, \tag{2}$$

$$Donation_i = \alpha + \beta_1 T1_i + \beta_2 T2_i + \beta_3 T3_i + \varepsilon_i. \tag{3}$$

The fair-fee model adjusts for the midpoint of the respondent's current-fee bracket; the donation model is unadjusted apart from treatment assignment. We use conventional OLS standard errors and apply Holm's correction separately within each outcome.

Study 1: invitation framing and survey take-up

The empathy-framed invitation did not increase survey take-up (difference = -0.24 pp, 95% CI [-1.54 pp, 1.06 pp], $p = .716$; odds ratio = 0.983, 95% CI [0.899, 1.076]).

Table 1: Study 1 invitation-letter take-up

Letter condition	N	Participants	Take-up rate
Efficiency letter	6,581	1,157	17.6%
Empathy letter	6,580	1,141	17.3%

Note: Empathy minus efficiency: -0.24 pp, 95% CI [-1.54 pp, 1.06 pp], $p = .716$; odds ratio 0.983.

Study 2: primary outcomes by treatment arm

Donation means were similar across arms; fair-fee means varied more but showed no common information-treatment pattern (Figure 2).

Table 2: Study 2 primary-outcome descriptives by treatment arm

Outcome	Arm	*N*	Mean	SD
Fair monthly fee proxy (EUR)	T0	475	86.49	97.79
	T1	524	80.24	99.49
	T2	482	83.85	102.88
	T3	508	90.53	104.67
Donation, explicit opt-outs = 0 (EUR)	T0	550	27.71	34.98
	T1	600	27.18	35.20
	T2	557	26.82	35.27
	T3	591	27.06	34.37

Note: The fair-fee outcome uses current-fee bracket midpoints when respondents report that their current fee is fair. Explicit lottery opt-outs are coded as EUR 0 for the primary donation outcome.

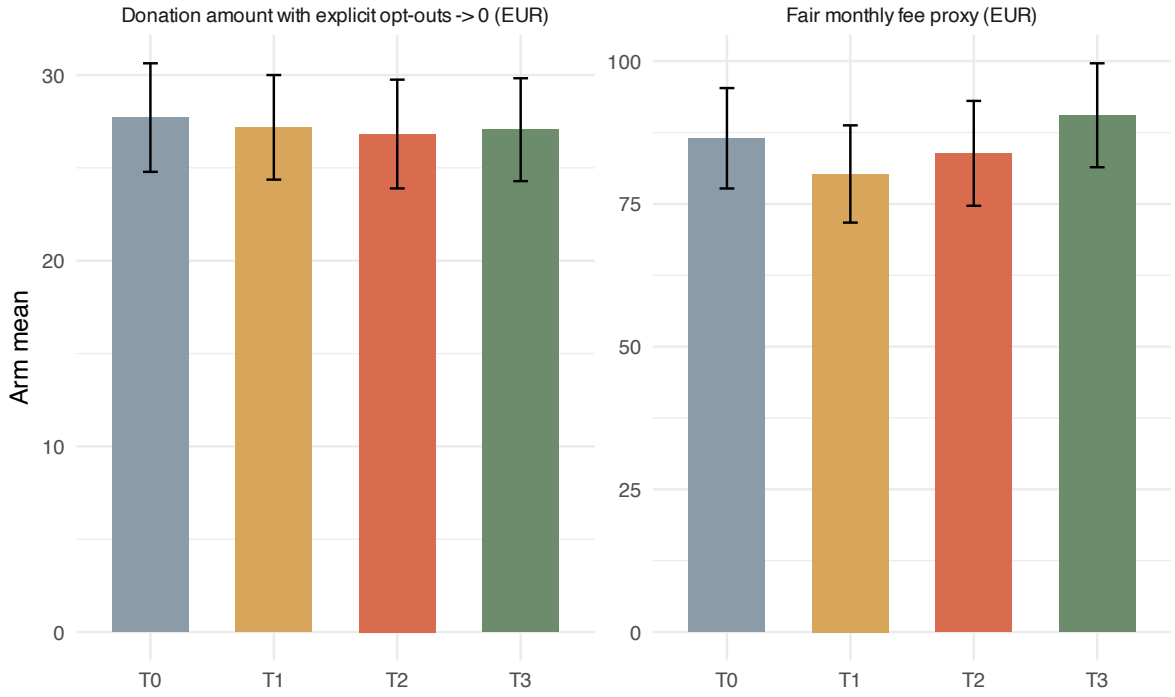


Figure 2: Primary outcomes by survey-treatment arm

Confirmatory survey-treatment contrasts

None of the six treatment-versus-control contrasts was distinguishable from zero after Holm adjustment (Table 3; Figure 3). HC2 standard errors and alternative fair-fee specifications did not change this conclusion (Appendix Table A1).

Table 3: Confirmatory Study 2 treatment contrasts

Outcome	Contrast	Estimate	95% CI	Raw p	Holm p
Fair monthly fee proxy (EUR)	T1 vs T0	-6.37	[-16.85, 4.11]	.233	.592
	T2 vs T0	1.65	[-9.03, 12.33]	.762	.762
	T3 vs T0	6.94	[-3.61, 17.50]	.197	.592
Donation, explicit opt-outs = 0 (EUR)	T1 vs T0	-0.53	[-4.57, 3.52]	.799	1.000
	T2 vs T0	-0.89	[-5.01, 3.23]	.673	1.000
	T3 vs T0	-0.65	[-4.71, 3.41]	.752	1.000

Note: OLS estimates with conventional model-based standard errors. Holm adjustment is applied separately to the three treatment-versus-control tests within each primary outcome.

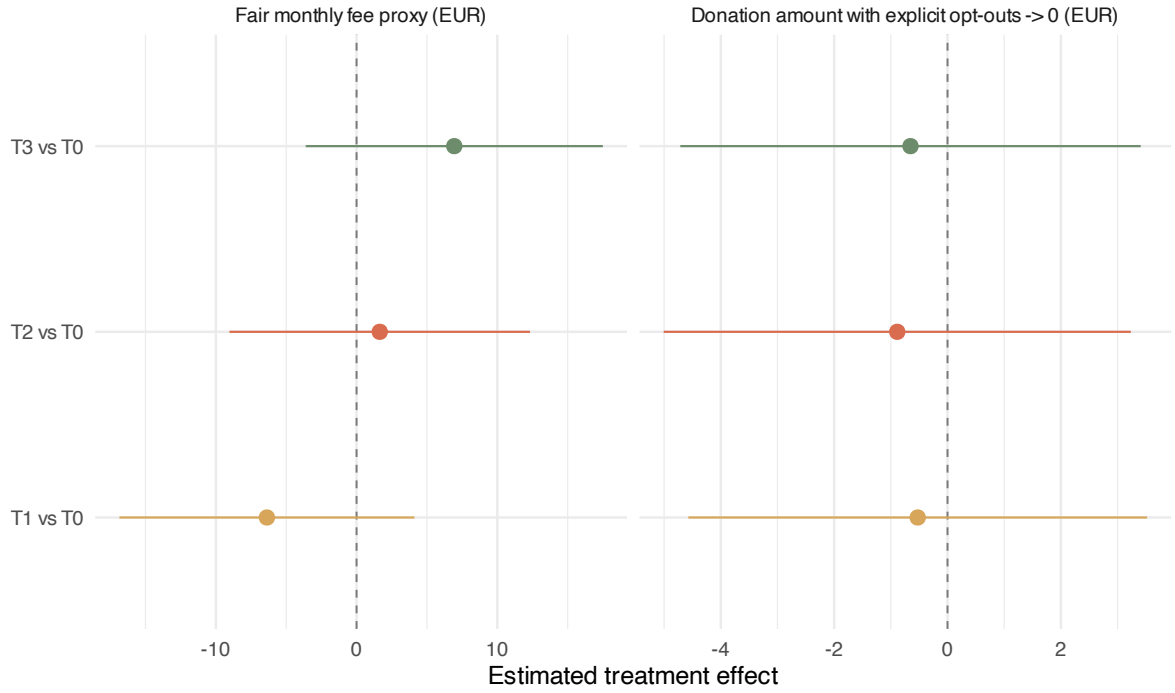


Figure 3: Confirmatory treatment effects relative to control

Focused secondary and exploratory results

The preregistered T2–T3 mechanism comparison was -5.29 EUR for fair fee (p = .322) and -0.23 EUR for donation (p = .910), offering no support for outcome-specific effects. Invitation-

letter interactions also showed no stable cumulative-empathy pattern (Appendix Table A3). The exploratory prior-belief, ideology, education, and income analyses were imprecise and did not alter the confirmatory conclusions (Appendix Tables A4–A7). Other secondary outcomes appear in Appendix Table A2.

Hypothesis assessment

Study 1 H1 was not supported. Study 2 H1a was directionally positive but not statistically supported after Holm adjustment, and H1b was not supported. H2 showed no clear evidence that survey-treatment effects were stronger after an empathy invitation. The exploratory moderation hypotheses H3 (ideology) and H4 (education) received no clear support; preregistered exploratory income moderation likewise showed no clear pattern. H5, the outcome-specific empathy-versus-fiscal mechanism, was not supported. The preregistration contains no H6; its unnumbered exploratory prior-belief-gap analysis also yielded no clear evidence.

Conclusion

The results provide little evidence that short empathy and information treatments increased solidarity in the METROPOLIS kindergarten-financing experiment. The empathy invitation did not increase take-up, and the survey treatments did not shift the fair-fee or donation outcomes after adjustment. The paper can therefore be developed as an informative null result about the limits of light-touch empathy and information interventions in a concrete local welfare-state setting.

The evidence points to a null result, but a useful one. Neither did empathy framing dincrease survey take-up in the field experiment, nor did the empathy-story, nor fiscal-information survey treatment produce clear confirmatory effects on fair-fee preferences or donation behavior. These null findings matter because the interventions were built around plausible mechanisms: beneficiary visibility, fiscal-information updating, and solidarity with families using public childcare (Batson et al., 1997; Haaland et al., 2023; van Oorschot, 2000). In this setting, those short messages were not enough.

One possible interpretation is that kindergarten-financing attitudes are relatively stable or only weakly connected to brief survey information. Another is that the treatments activated countervailing considerations. Fiscal information may increase awareness of public support while also making the taxpayer burden salient. Empathy stories may increase concern for beneficiaries among some respondents but produce skepticism or reactance among others. With the current aggregate estimates, we cannot fully separate these possibilities.

Another possible interpretation of the null effect may be a ceiling effect in parental group identity in the German welfare state. Being a parent of small children and being embedded in a web of public services and money transfers may be such a dominant way of viewing oneself that people are insensitive to information about others who fall into the same category. In experimental terms, the control group may already be near the ceiling in terms of feeling and "being" kindergarten parents.

One piece of evidence was interesting. Respondents on average thought parents paid a much larger share of kindergarten costs than the treatment benchmark implied. Yet correcting that context did not produce the expected confirmatory effects. This is useful for theories of information effects

because it shows that even large factual gaps do not automatically translate into attitude or behavior change.

product of the welfare state, socialisation, growing up with welfare stats, welfare state attitudes Is it about the group that we studied or is it about the programme that we studied? parents, small kids,

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Supplementary analyses

This appendix reports the sensitivity, secondary-outcome, and moderation estimates referenced in the main Results section. Unless noted otherwise, estimates use conventional model-based standard errors and unadjusted two-sided tests.

Table A1: Alternative specifications for the primary Study 2 outcomes

Outcome	Comparison	Estimate	SE	p	N
Stated fair fee among fairness1 != 0	T1 vs T0	1.49	9.41	.874	834
	T2 vs T0	3.85	9.93	.699	834
	T3 vs T0	9.36	9.74	.337	834
Fair-fee proxy with fee-bracket fixed effects	T1 vs T0	-5.21	5.32	.328	1,957
	T2 vs T0	2.35	5.42	.665	1,957
	T3 vs T0	7.79	5.36	.146	1,957
Fair monthly fee proxy (EUR); HC2	T1 vs T0	-6.37	5.37	.236	1,957
	T2 vs T0	1.65	5.38	.759	1,957
	T3 vs T0	6.94	5.45	.203	1,957
Donation, explicit opt-outs = 0 (EUR); HC2	T1 vs T0	-0.53	2.07	.800	2,298
	T2 vs T0	-0.89	2.11	.675	2,298
	T3 vs T0	-0.65	2.06	.751	2,298

Note: Treatment contrasts are relative to T0. HC2 rows use heteroskedasticity-consistent standard errors; other rows use conventional model-based standard errors. The stated-only outcome excludes respondents who considered their current fee fair.

Table A2: Secondary Study 2 outcomes

Outcome	Comparison	Estimate	SE	p	N
Donation among confirmed participants (EUR)	T1 vs T0	-0.30	2.22	.894	2,040
	T2 vs T0	-0.11	2.27	.961	2,040
	T3 vs T0	0.02	2.24	.991	2,040
Financing satisfaction (1-11)	T1 vs T0	-0.23	0.18	.210	2,080
	T2 vs T0	0.02	0.19	.900	2,080
	T3 vs T0	0.29	0.18	.116	2,080
Lottery opt-out (log odds)	T1 vs T0	0.09	0.19	.636	2,298
	T2 vs T0	0.26	0.19	.175	2,298
	T3 vs T0	0.23	0.19	.236	2,298

Note: Treatment contrasts are relative to T0 and are unadjusted for multiple testing. Lottery opt-out estimates are log-odds coefficients; other estimates are OLS coefficients in the displayed units.

Table A3: Interactions between survey treatment and invitation-letter framing

Outcome	Comparison	Estimate	SE	p	N
Fair monthly fee proxy (EUR)	T1 interaction with empathy letter	-36.66	19.92	.066	619
	T2 interaction with empathy letter	-23.35	19.59	.234	619
	T3 interaction with empathy letter	-23.57	19.80	.234	619
Donation (EUR)	T1 interaction with empathy letter	-12.91	7.50	.086	733
	T2 interaction with empathy letter	-3.60	7.37	.625	733
	T3 interaction with empathy letter	-4.25	7.44	.568	733

Note: Each coefficient is the change in the treatment-versus-T0 contrast under the empathy rather than efficiency invitation letter. The fair-fee model adjusts for current-fee midpoint.

Table A4: Prior-belief-gap associations within the fiscal-information arm

Outcome	Comparison	Estimate	SE	p	N
Fair monthly fee proxy (EUR)	10-percentage-point larger absolute belief gap	-2.66	1.72	.121	451
Donation (EUR)	10-percentage-point larger absolute belief gap	-0.84	0.63	.181	512

Note: Models are restricted to T3. Estimates are exploratory, non-causal associations with a 10-percentage-point larger absolute prior-belief gap.

Table A5: Treatment-effect moderation by political ideology

Outcome	Comparison	Estimate	SE	p	N
Fair monthly fee proxy (EUR)	T1 interaction (one-SD increase)	-10.55	6.28	.093	1,634
	T2 interaction (one-SD increase)	-4.44	6.60	.501	1,634
	T3 interaction (one-SD increase)	-3.65	6.32	.564	1,634
Donation (EUR)	T1 interaction (one-SD increase)	-3.05	2.35	.195	1,852
	T2 interaction (one-SD increase)	-0.61	2.44	.801	1,852
	T3 interaction (one-SD increase)	-2.02	2.31	.382	1,852

Note: The moderator is standardized. Each coefficient shows how the treatment-versus-T0 contrast changes with a one-standard-deviation increase. Fair-fee models adjust for current-fee midpoint. Tests are unadjusted for multiplicity.

Table A6: Treatment-effect moderation by education

Outcome	Comparison	Estimate	SE	p	N
Fair monthly fee proxy (EUR)	T1 interaction (one-SD increase)	-1.26	5.81	.828	1,852
	T2 interaction (one-SD increase)	-2.02	5.95	.735	1,852
	T3 interaction (one-SD increase)	-2.43	5.74	.672	1,852
Donation (EUR)	T1 interaction (one-SD increase)	-2.49	2.18	.252	2,137
	T2 interaction (one-SD increase)	-1.48	2.23	.506	2,137
	T3 interaction (one-SD increase)	-3.21	2.13	.132	2,137

Note: The moderator is standardized. Each coefficient shows how the treatment-versus-T0 contrast changes with a one-standard-deviation increase. Fair-fee models adjust for current-fee midpoint. Tests are unadjusted for multiplicity.

Table A7: Treatment-effect moderation by household income

Outcome	Comparison	Estimate	SE	p	N
Fair monthly fee proxy (EUR)	T1 interaction (one-SD increase)	-5.34	5.89	.365	1,775
	T2 interaction (one-SD increase)	-8.14	5.95	.172	1,775
	T3 interaction (one-SD increase)	1.73	5.96	.772	1,775
Donation (EUR)	T1 interaction (one-SD increase)	-3.37	2.20	.126	2,042
	T2 interaction (one-SD increase)	-2.79	2.21	.206	2,042
	T3 interaction (one-SD increase)	-1.85	2.16	.391	2,042

Note: The moderator is standardized. Each coefficient shows how the treatment-versus-T0 contrast changes with a one-standard-deviation increase. Fair-fee models adjust for current-fee midpoint. Tests are unadjusted for multiplicity.

Supplementary Appendix

Information, Empathy, and In-Group Solidarity in Kindergarten Financing: A Field and Survey Experiment in METROPOLIS

2026-10-16

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AUTHOR-REVIEW DRAFT. This document assembles the available design evidence and identifies the remaining documentation. It is not a submission-ready supplement. Bracketed author actions must be resolved before submission. German source passages are reproduced from local project documents. English translations are working translations prepared for this draft. Statements about the fielded design refer to the available 16 April 2026 instrument export unless another source is named. An export shows stored settings. It does not establish that those settings were unchanged throughout fieldwork.

The paper studies an invitation-letter experiment and a four-arm survey experiment. Later focus groups and Wave 2 are outside its present scope. Their existence is disclosed where participants were invited to those activities. Their results are not added to this paper as if they were part of its original design.

A. Setting, design, and populations

A.1 Research question and setting

The study examines whether brief appeals to empathy and information about collective financing affect expressed support for solidarity among parents of kindergarten children in METROPOLIS, Germany. The setting links household contributions to a service financed largely through public funds. Parents face different contributions. This creates a distinction between the amount a household currently pays, its judgment of a fair contribution, and its willingness to give up a possible lottery gain for a local organization supporting children.

The municipality supplied the invitation frame and sent the invitation. The survey ran online using Unipark/EFS. The stored project requirements identify EFS version 24.1. The instrument covers kindergartens under municipal and other organizational providers. The population should therefore not be described as users of municipally operated kindergartens alone.

[**AUTHOR ACTION A1**] Supply the relevant fee schedule, its effective date, the administrative definition of the eligible frame, and the source and reference year for the financing figures. The stimulus wording is evidence of what the screen said. It is not independent verification of municipal accounts.

A.2 Two sequential experiments

Study 1 compares an efficiency-oriented invitation with an empathy/community-oriented invitation. Assignment occurs before survey entry. The experimental unit is a retained family identifier in the mailing frame. One selected addressee receives the assigned invitation.

Study 2 compares no additional screen with three screens: generic kindergarten information, an empathy narrative, and fiscal/community information. Its recorded assignment variable is `c_0001`. The unit is the survey participant record. The two experiments have different denominators. Study 1 includes randomized invitation records. Study 2 outcomes are available for retained survey respondents with usable measurements.

The second experiment does not identify the average effect of information among all invited families. People must enter the survey to encounter its materials. Comparisons of survey-treatment arms can be interpreted causally under the survey randomization and appropriate assumptions about missing outcomes. Comparisons involving invitation assignment among survey completers require additional care. Survey participation occurs after invitation assignment and may depend on it.

A.3 Hypotheses and analysis labels

The local plan specifies that the empathy invitation increases survey take-up. For Study 2, its primary expectation is that the empathy and fiscal treatments increase fair-fee preferences and donation amounts relative to control. The generic-information arm provides a further benchmark.

The plan also expects stronger survey-treatment effects following the empathy invitation. It labels ideology and education moderation as exploratory. Its final section states an exploratory mechanism expectation: the empathy narrative should have relatively stronger donation effects, while fiscal information should have relatively stronger fair-fee effects. Income, area deprivation if linkable, prior beliefs, baseline fees, and perceived unfairness direction appear as additional exploratory checks.

The manuscript separates the two outcomes under Study 2 H1 into H1a and H1b. This is an expositional split. It must not be presented as a separate registration. The local plan has no H6.

[**AUTHOR ACTION A2**] Verify these labels against the final timestamped registration. The local text in Part N is the comparison document used for this draft.

B. Sampling frame, recruitment, and invitation assignment

B.1 Frame construction

The source workbook contains 15,788 rows. The documented preparation removes 475 rows with blank family identifiers and 2,152 repeated rows beyond the first instance of a nonblank identifier. The retained frame has 13,161 distinct identifiers:

$$15{,}788 - 475 - 2{,}152 = 13{,}161.$$

There are 2,039 duplicated nonblank identifiers before retaining the first record. The number of duplicated identifiers is distinct from the number of extra rows removed. Frame cleaning precedes assignment. It is not post-treatment exclusion.

The canonical script trims family identifiers and retains the first row for each nonblank identifier. It does not sample one record from duplicate sets at random. The final account should explain why first-row retention gives the intended invitation unit and whether repeated rows reflect siblings or another administrative structure. This draft does not infer the cause of repetition.

B.2 Addressee selection and message assignment

The frame distinguishes two potential addressees, ZPF1 and ZPF2. In 11,020 retained records, the script marks a second addressee as available on the basis of a nonblank ZPF2 gender field. In 2,141 records, it does not. Where a second addressee is available, half of the records are assigned to ZPF2 and half to ZPF1. Records without ZPF2 go to ZPF1. This produces 7,651 ZPF1 addressees and 5,510 ZPF2 addressees.

The script sets seed 2026 once. Its random operations generate unique survey codes first, select addressee targets second, and shuffle message assignments third. It shuffles within addressee target and alternates the two messages. Thus, message assignment is count-balanced within target. The script does not stratify message assignment on the full set of household characteristics. Its fixed random-operation order and the source row order matter for exact reproduction.

The resulting invitation arms contain 6,581 efficiency invitations and 6,580 empathy invitations. Addressee selection and message assignment are separate operations. Neither should be described as the other.

B.3 Unique codes and delivery

Each retained record receives a six-character code. The code generator excludes selected ambiguous letters and uses a restricted character set. The code connects the invitation, survey access, and participant-status records. It is a linkage identifier and should not be included in a public appendix or public replication file.

The exact experimental message paragraphs appear in Part D. The final printed letter should also be archived with personal details, unique codes, and identifying review information removed as appropriate. The experimental paragraphs alone do not document the envelope, sender, common instructions, QR code, access procedure, or any common incentive information.

[**AUTHOR ACTION B1**] Confirm the dispatch date or dates, postal method, final printed template, delivery failures, reminders, survey-opening and closing dates, and any added mailing batches. A 432-record leftover workflow exists in the archive. Establish whether it was dispatched and whether it belongs in this paper's randomized population. Do not add it merely because a preparation file exists.

B.4 Recruitment outcome

The existing analysis joins the 13,161 invitation records to status data by unique code. It codes statuses 31, 32, 33, and 34 as take-up and other statuses as zero. The available retained survey data contain statuses 31 and 32, labelled completed and completed after break.

The local plan defines take-up as accessing the survey and providing consent. The implemented status rule must be reconciled with that definition. Completion-based status does not by itself identify all people who accessed the survey or consented and then stopped.

Assigned invitation	Randomized records	Status-defined participants	Remaining records	Status-defined participation
Efficiency	6,581	1,157	5,424	17.58%
Empathy	6,580	1,141	5,439	17.34%
Total	13,161	2,298	10,863	17.46%

These are frame-based participation proportions under the existing code. They should not be labelled an AAPOR response rate without documenting the relevant disposition categories and denominator. They do not establish the number of delivered letters.

C. Survey assignment, routing, and exposure

C.1 Assignment crosswalk

Recorded `c_0001`	Analysis label	Substantive condition	Stored route
1	T0	No additional treatment screen	Bypasses the three intervention screens
4	T1	Generic information / thematic prime	Filter 7601926 to page 7571738
2	T2	Empathy narrative	Filter 7601924 to page 7571736
3	T3	Fiscal/community information	Filter 7601925 to page 7576345

The recorded numbers are not a sequence of substantive labels. For example, code 4 represents T1. A replication script must use the explicit crosswalk rather than subtracting one from the recorded number.

C.2 Timing and allocation procedure

The GPX places the `treatment_id` trigger after the welcome page, W1, page 7599681. The trigger draws an integer from 1 to 4 into `c_0001`. Its stored setting is `balanced=0`. It is upstream of the consent pages and intervention screens. The local plan describes assignment among consenting participants. This difference should be reported after checking deployment history.

Unequal numbers among retained completions do not establish a randomizer failure. They may reflect random variation, differential retention, or both. Conversely, those retained numbers cannot establish equal initial allocation. To document attrition, the final package

needs all initially assigned records and their subsequent dispositions by arm.

C.3 Order and competing exposures

The instrument presents the welcome and data-use consent material, questions about the household and child, contextual and political measures, current fees, and beliefs about financing before the intervention pages. A content-recognition question follows the intervention route. Fairness and financing-satisfaction measures follow it. Sociodemographic items and invitations to further study components precede the lottery allocation and contact procedure.

The control group encounters the same surrounding questionnaire without an additional treatment screen. Thus, the experimental comparison is conditional on this questionnaire context. Earlier questions may already direct attention to fees, financing, and shared provision.

The three intervention screens have different content and different images. The experiment estimates the effects of these screen packages. It does not separately identify the effect of empathy wording, fiscal numbers, an image, or reading time.

C.4 Recognition, fidelity, and blinding

The recognition item, `v_57`, asks what the text was mainly about. Its options concern kindergarten facts, a family's experience, taxpayer financing, a daily routine, and seasonal activities. The expected answers for T1, T2, and T3 are 1, 2, and 3. The stored filter and exact item appear in the instrument extracts in Part M.

Recognition is measured after assignment. It must not be used to exclude respondents from the primary assigned-arm analysis. Report its response distribution by assigned arm as a descriptive delivery/comprehension check. Passing it does not establish emotional engagement or belief updating.

Participants can see the material assigned to them. There is no evidence here of a tested blinding procedure. The survey standardizes screen delivery, but this is not proof that researchers were blind during preparation or analysis. No practice rounds or repeated within-person experimental treatment sequence are part of these two experiments.

[**AUTHOR ACTION C1**] Supply the deployed randomizer history, page-display records where available, device screenshots, and any minimum-display or back-button rules relevant to exposure. Confirm whether a participant could revisit or change an assignment. Do not infer exposure from completion alone.

D. Original experimental materials

D.1 Invitation paragraphs in German

The following paragraphs match the final-message document dated 11 March 2026, 14:45, and the canonical randomization script. Lowercase openings are retained from the source. Common letter material remains to be attached.

Efficiency invitation

> wie ist das Leben mit Kindergarten-Kindern in METROPOLIS? Helfen Sie uns bitte, dies besser zu verstehen! Ihre Antworten tragen dazu bei, die Qualität und Effizienz von öffentlichen Dienstleistungen in METROPOLIS zu verbessern. Sie unterstützen mit Ihrer Teilnahme an der Umfrage die Weiterentwicklung der städtischen Einrichtungen.

Empathy/community invitation

> wie ist das Leben mit Kindergarten-Kindern in METROPOLIS? Helfen Sie uns bitte, dies besser zu verstehen! Mithilfe Ihrer Eindrücke können wir METROPOLIS noch besser für Familien mit Kindern machen. Sie unterstützen mit Ihrer Teilnahme an der Umfrage die Gemeinschaft von Eltern mit Kindergartenkindern in METROPOLIS

D.2 Working English translations of invitation paragraphs

Efficiency invitation. What is life like with kindergarten children in METROPOLIS? Please help us understand it better! Your answers help improve the quality and efficiency of public services in METROPOLIS. By taking part in the survey, you support the further development of the city's facilities.

Empathy/community invitation. What is life like with kindergarten children in METROPOLIS? Please help us understand it better! Your impressions can help us make METROPOLIS an even better place for families with children. By taking part in the survey, you support the community of METROPOLIS parents with kindergarten children.

D.3 Original survey screens and images

The next sections reproduce text recovered directly from the GPX. The associated images are extracted from its embedded image assets. They are not screenshots of a browser display. Original paragraph and heading order is retained, while this document changes typography. This distinguishes the actual German instrument from the separate English treatment draft, which contains different narrative and fiscal details.

T1: generic information

Source: GPX page 7571738.

> Kindergärten in METROPOLIS
>
> Bitte lesen Sie aufmerksam den folgenden Text.

In METROPOLIS gibt es aktuell über 200 Kindergärten. Etwas mehr als ein Drittel davon werden von der Stadt METROPOLIS selbst betrieben, der Rest der Kindergärten ist in der Trägerschaft anderer Organisationen.

Die Kindergärten verteilen sich über das gesamte Stadtgebiet, vom äußersten Norden bis zum äußersten Süden von METROPOLIS. Der kleinste Kindergarten hat gerade einmal 19 Plätze und eine Gruppe, der größte Kindergarten 177 Plätze in 8 Gruppen.

Die Betreuungszeiten unterscheiden sich zwischen den Kindergärten. Einige Kindergärten bieten eine reine Vormittagsbetreuung an, andere öffnen ganztägig.

Eltern suchen Betreuungsplätze über das zentrale Online-Portal Kita-Place. Dieses System erfasst alle Standorte, unabhängig von der Trägerschaft. Das Jugendamt von METROPOLIS aktualisiert das Verzeichnis der Einrichtungen vor Beginn jedes neuen Kindergartenjahres.



Figure 1: Embedded image associated with T1: generic information.

T2: empathy narrative

Source: GPX page 7571736.

Sofia kann in den Kindergarten gehen

Bitte lesen Sie aufmerksam den folgenden Text.

Julia und Daniel leben mit ihrer dreijährigen Tochter Sofia in METROPOLIS. Julia arbeitet als Sekretärin, Daniel als Lagerarbeiter. Obwohl beide viel arbeiten, ist das Geld oft knapp.

In METROPOLIS zahlen Familien mit geringerem Einkommen niedrigere

Kindergarten-Beiträge. Nur deshalb können Julia und Daniel es sich leisten, dass Sofia den Kindergarten besucht.

Die Mutter Julia: „Als die Zusage für den Kindergarten kam, war ich so erleichtert. Wir hatten so gehofft, dass es klappt. Sofia geht total gerne hin und hat dort schon einige Freundinnen aus dem Viertel. Wir hatten schon ein bisschen Sorge, dass wir am Ende mehr zahlen müssten. Dann hätten wir uns Sofias Kindergarten wahrscheinlich nicht leisten können. So fühlt es sich jetzt einfach richtig gut an.“



Figure 2: Embedded image associated with T2: empathy narrative.

T3: fiscal/community information

Source: GPX page 7576345.

Gemeinsam für Kinder

Bitte lesen Sie aufmerksam den folgenden Text.

Als Eltern eines Kindergartenkindes sind Sie Teil einer starken Gemeinschaft, deren Mitglieder sich gegenseitig helfen.

Ein Kindergartenplatz in METROPOLIS kostet monatlich rund 960 Euro. Eltern zahlen davon zwischen 0 und rund 500 Euro – im Durchschnitt 40 Euro. Den Großteil von etwa 920 Euro tragen alle Steuerzahler in Nordrhein-Westfalen (NRW) gemeinsam. Selbst beim höchsten Elternbeitrag übernehmen die Steuerzahler rund die Hälfte der Kosten.

Innerhalb der Gemeinschaft werden die finanziellen Kosten umverteilt. Alle Bürger, ob mit oder ohne Kinder, finanzieren die Kinderbetreuung gemeinsam. Sie unterstützen Ihre Kinder, so wie Sie durch Ihren Elternbeitrag und als Steuerzahler andere Kinder unterstützen.

Jedes Mitglied der Gemeinschaft ermöglicht die Betreuung von Kindergartenkindern in METROPOLIS mit seinem Beitrag: als Steuerzahler in NRW oder als Beitragszahler in METROPOLIS.

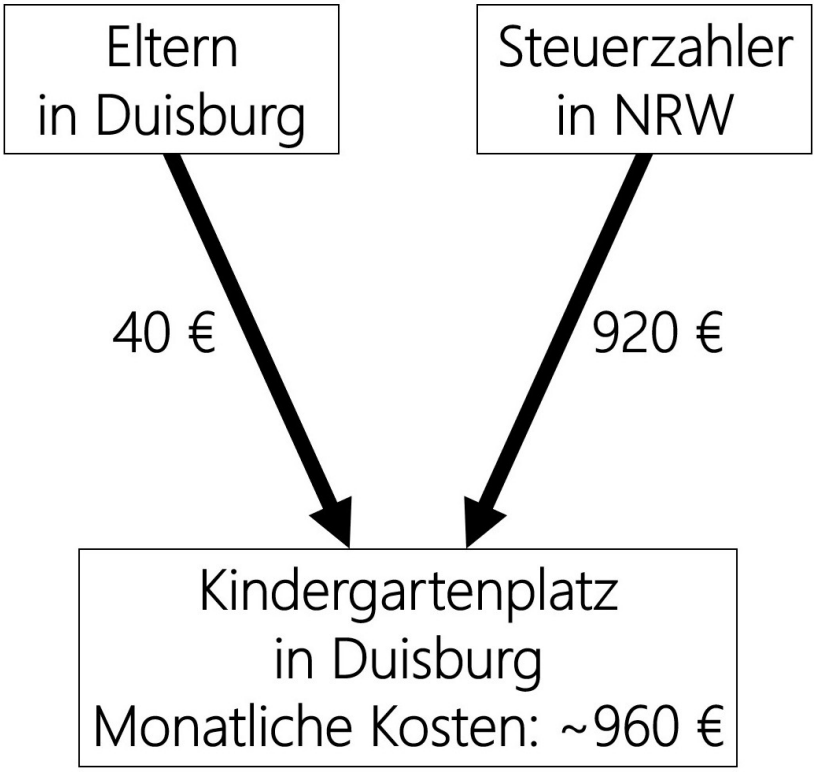


Figure 3: Embedded image associated with T3: fiscal/community information.

D.4 Working English translations of survey screens

T1: Kindergartens in METROPOLIS

Please read the following text carefully.

METROPOLIS currently has more than 200 kindergartens. A little more than one third are operated by the city itself. The remaining kindergartens are run by other organizations.

TThe kindergartens are spread across the city, from the far north to the far south of METROPOLIS. The smallest kindergarten has just 19 places and one group. The largest has 177 places in eight groups.

Opening hours vary between kindergartens. Some offer morning care only. Others are open all day.

Parents look for childcare places through the central online portal Kita-Place. This system lists all locations, regardless of provider. METROPOLIS's youth welfare office updates the directory of institutions before the start of each new kindergarten year.

T2: Sofia can go to kindergarten

Please read the following text carefully.

Julia and Daniel live in METROPOLIS with their three-year-old daughter Sofia. Julia works as a secretary. Daniel is a warehouse worker. Although both work a lot, money is often tight.

In METROPOLIS, families with lower incomes pay lower kindergarten contributions. Only because of this can Julia and Daniel afford for Sofia to attend kindergarten.

Sofia's mother Julia says: "When the offer of a kindergarten place came, I was so relieved. We had really hoped it would work out. Sofia loves going there and has already made some friends from the neighborhood. We were a little worried that we might end up having to pay more. Then we probably could not have afforded Sofia's kindergarten. As it is, it simply feels really good."

T3: Together for children

Please read the following text carefully.

As the parent of a kindergarten child, you are part of a strong community whose members help one another.

A kindergarten place in METROPOLIS costs around EUR 960 per month. Parents pay between zero and around EUR 500 of this, with an average of EUR 40. All taxpayers in North Rhine-Westphalia jointly cover most of the cost, around EUR 920. Even at the highest parental contribution, taxpayers cover around half the cost.

Financial costs are redistributed within the community. All citizens, with or without children, jointly finance childcare. They support your children, just as you support other children through your parental contribution and as a taxpayer.

Every member of the community makes childcare for METROPOLIS's kindergarten children possible through their contribution: as a taxpayer in North Rhine-Westphalia or as a contributor in METROPOLIS.

D.5 Material-version discrepancy

The separate English document, `field_experiment_treatments.docx`, describes a different family narrative and a science-based evidence treatment. Its contribution treatment states EUR 993 total costs, EUR 70 parental payments, and EUR 923 taxpayer payments. The generic German screen in the GPX is descriptive information about local kindergartens. It is not the science-based screen in that English document.

The appendix uses the GPX as the available instrument snapshot. Do not silently combine the English draft with the German screens. [**AUTHOR ACTION D1**] Confirm which versions were deployed throughout fieldwork. Attach final print and browser screenshots. Confirm image provenance and reproduction rights before public deposit.

E. Outcomes, covariates, and coding

E.1 Measurement map

Construct	Source field	Rule in the existing analysis
Survey arm	`c_0001`	1=T0, 4=T1, 2=T2, 3=T3
Consent	`v_1`, `v_126`	Initial agreement or agreement at confirmation
Current monthly contribution	`v_180`	Bracket representatives in EUR
Fairness of current contribution	`v_65`	Recorded 1-11 transformed by subtracting 6
Stated fair monthly contribution	`v_67`	Nonnegative numeric amount where routed
Financing satisfaction	`v_68`	Valid responses 1-11
Parent-share estimate	`v_56`	Responses between 0% and 100%
Treatment recognition	`v_57`	Descriptive recognition measure
Lottery donation	`v_168`	Codes 1-11 correspond to EUR 0-100
Lottery contact choice	`v_116`	Codes 1-3 indicate a selected contact procedure
Ideology	`v_37`	Valid responses 1-11; higher values further right
Education	`v_76`	Codes 1-8 retained; other qualification code handled separately
Household income	`v_95`	Five ordered categories

Exact German item wording and response categories appear in Part M. These are measurement rules documented from the current code. Their presence here does not certify that every rule matches the final registration.

E.2 Current contribution and the fair-fee outcome

The questionnaire records current fees in brackets. The existing analysis uses the following representatives:

Recorded category	Representative in EUR
1	0
2	25
3	75
4	125
5	175
6	225
7	275
8	325
9	375
10	425
11	475

These are approximate representatives. Where the printed bounds are consecutive integers, they are not the exact arithmetic midpoints. The resulting outcome should therefore be called a fair-fee proxy, rather than an exact fee amount for every respondent.

Let C_i denote the current-fee representative, F_i the transformed fairness item, and S_i the stated fair fee. The outcome is

$$Y_i^{fee} = \begin{cases} C_i & F_i = 0, \\ S_i & F_i \neq 0, \\ \text{missing} & \text{otherwise.} \end{cases}$$

For respondents who regard their current contribution as fair, the current contribution enters

the dependent variable directly. The local plan therefore adjusts fair-fee regressions for the current contribution. A usable outcome alone does not guarantee inclusion in that regression. Respondents with a stated fair fee but missing current-fee category are excluded from its complete-case model frame.

The German fairness item anchors -5 at "unfairly too low" and +5 at "unfairly too high." The local preregistration prose reverses these endpoints. The zero/nonzero branch is unaffected by that reversal. Directional interpretations, secondary fairness outcomes, and direction-based subgroups are affected.

[**AUTHOR ACTION E1**] Resolve the scale description and audit all direction-based analyses. Confirm handling of the blank category 12 shown in the codebook. The existing fairness transformation has no explicit 1-11 range check before subtracting six. Verify whether any retained record uses that blank category.

E.3 Donation decision and incentive structure

The survey offers a chance to win one of 25 prizes worth EUR 100 each. Respondents choose how much of a possible EUR 100 prize to give to [LOCAL CHILDREN'S ORGANISATION] Choices range from EUR 0 to EUR 100 in EUR 10 steps. The remaining amount is described as a digital gift voucher. This is an allocation contingent on winning. It is not an immediate payment from every respondent's own funds.

For valid allocation code $a_i \in \{1, \ldots, 11\}$, the selected donation is $10(a_i - 1)$ euros. The existing code defines a confirmed lottery participant as a person with a valid allocation and a selected contact option, `v_116` in 1-3. An explicit refusal at either the allocation or contact

item is assigned EUR 0 for the primary outcome. Other unresolved missing responses remain missing.

The code checks the confirmed-participant branch before the explicit-opt-out branch. The final audit should test for contradictory responses satisfying both branches and state the chosen precedence. Contact-option selection is a proxy for participation. It does not itself verify that a valid address or number was supplied or that a prize was delivered.

The original April export supports 2,040 confirmed-participant observations and 258 explicit opt-outs, yielding 2,298 primary donation observations under the stated zero-coding rule. There are 2,198 usable allocation-item amounts. The remaining 100 respondents explicitly refuse at that item. Another 158 select a donation amount and then refuse at the contact step.

The distinction between a selected EUR 0 donation and opting out of the lottery must be retained in the raw and prepared data. Both may contribute zero to the main outcome, but they represent different decisions. The conditional-on-participation analysis is secondary. Participation can itself respond to treatment, so that conditional comparison does not automatically preserve randomization.

E.4 Secondary measures and moderators

Financing satisfaction is recorded from 1, extremely dissatisfied, to 11, extremely satisfied. Fairness perceptions are a separate secondary outcome. The direct stated fair-fee amount, among respondents routed to it, is an alternative outcome definition. It changes the sample and should not replace the registered composite without disclosure.

Ideology, education, and income are standardized in the existing moderation code using the retained survey sample's mean and standard deviation. Education and income are entered as ordered numeric categories. This imposes equal spacing between category codes. The appendix should state that assumption. Any categorical specification added later should be labelled as a sensitivity analysis.

The fiscal belief gap is the absolute difference between the estimated parent-financed share and the benchmark. The existing code uses $100(70/993) = 7.05\%$. The stored fiscal screen implies $100(40/960) = 4.17\%$. These are different moderators. The final analysis should use the verified fielded benchmark and retain the old specification as a documented sensitivity if useful. Do not reinterpret an old estimate under a new benchmark without recomputation.

Within-T3 regressions relate outcomes to the size of a prior-belief gap. They are associations within one assigned arm. They do not identify a treatment-by-gap effect relative to an untreated counterfactual. No measured post-treatment emotional response is established here as a causal mediator.

E.5 Missing values and exclusions

The existing cleaning function converts numeric values 88, 99, 888, 999, -66, -77, -88, and -99 to missing across numeric fields. It also handles selected character sentinels. A blanket rule can remove plausible euro amounts, including EUR 88 or EUR 99, if applied to an unrestricted fee response. A final variable-specific audit must distinguish substantive amounts from item-specific nonresponse codes.

The original analysis saves raw copies of donation and contact responses before cleaning. That preserves explicit opt-outs. The merged RDS already replaces those codes with missing values. Saving a raw copy after reading that RDS cannot recover the distinction.

The current preparation keeps non-test records with status 31 or 32. It does not explicitly filter on consent in the analyzed chunk. The separate research-design audit checks that retained records satisfy the consent rule. These are distinct implementation facts. The final preparation should make the inclusion rule and its verification explicit.

No outcome-based trimming or winsorization is established as part of the local plan. It permits clearly impossible values to be treated as missing. It does not justify dropping unusually large but feasible values. Do not add speed, quality-score, or recognition exclusions to the confirmatory sample without documenting a departure.

F. Participant flow and data versions

F.1 Observed and model-specific samples

The September merged snapshot contains 13,163 wide records and 2,298 flagged Wave 1 survey records. The invitation experiment has 13,161 randomized records. The two-record difference requires a construction-level explanation. It is not evidence of two additional randomized invitations.

The following counts were reproduced on 16 October 2026 using the selected merged file and the preparation chunk used by `research_design.qmd`:

Arm	Retained survey records	Fair-fee outcome observed	Adjusted fair-fee model	Donation after merged-file preparation
T0	550	475	465	496
T1	600	524	515	536
T2	557	482	477	488
T3	591	508	500	520
Total	2,298	1,989	1,957	2,040

Thirty-two observations have a usable fair-fee outcome but lack the current-fee covariate. There are 309 retained respondents without an observed fair-fee proxy and 341 outside the adjusted fair-fee model.

The last column is a data-version diagnostic, not the final preregistered donation sample. It excludes opt-outs whose original response codes have become missing. The original export supports 2,298 primary donation observations after retaining those codes and applying the intended zero coding.

F.2 Reconciliation of the donation denominator

The original April export has the same 2,298 retained respondent IDs as the merged Wave 1 subset. Among those matched IDs, 100 original allocation refusals and 158 original contact refusals are missing in the corresponding merged fields. This accounts for all 258 lost explicit-opt-out records under the current preparation. It reconciles the difference between the two donation denominators. It does not validate every other transformation in the merged pipeline.

The final replication source must retain a pre-cleaning opt-out indicator or reproduce it from the original export using a verified key. Zero-filling every missing merged response would not be an acceptable substitute. Future exports might contain genuine missingness unrelated to opting out.

F.3 Linkage of the two experiments

All 2,298 retained survey records have distinct canonical `code` values in the merged file that match invitation codes in the mailing list. None of those canonical codes is missing or duplicated. Under the legacy procedure that extracts a code from `c_0002` or reconstructed `invite_email`, only 733 records match.

This supports complete structural linkage through the canonical merged key. The final package still needs the code that constructs that key. A matching identifier is stronger evidence than equal aggregate counts, but the merge history must establish that the identifier was assigned to the correct survey record.

The final linked-model sample is further limited by each outcome and any required covariates. Do not report 2,298 as the sample size for every letter-by-survey interaction model.

F.4 What remains unavailable for a complete flow diagram

Retained completions comprise 2,220 status-31 records and 78 status-32 records. These counts do not reveal all visitors assigned by the welcome-page randomizer. A final flow diagram should show, for each arm, initial assignment, consent, screen display, completion, usable outcome, and model inclusion. Delivery failures and exclusions before letter randomization belong in the Study 1 portion.

[**AUTHOR ACTION F1**] Obtain the all-disposition survey export and deployment log. Report unknown flow cells as unknown until verified. Do not infer zero attrition or complete exposure from the completion-only subset.

G. Preregistration, implementation, and departures

G.1 Registration status

Local preregistration drafts exist, including a dated Word file and `field_exp_code/prereg.md`. Their contents describe an intended two-stage design. A final public or anonymized registration URL, registration identifier, timestamp, and immutable registered version have not been established in this review.

The supplementary file in Part N is therefore described as a local preregistration text. It must not be called a verified pre-data registration. Confirmatory labels in this draft reflect

that text and the current manuscript. They remain subject to the timestamped record.

G.2 Plan-to-implementation comparison

Topic	Local plan	Available implementation evidence	Required resolution
Letter population	Approximately 16,000 eligible households	13,161 retained randomized IDs after frame cleaning	Explain planning estimate and final frame
Take-up	Access plus consent	Statuses 31-34 coded as participation	Verify disposition definitions and outcome alignment
Survey assignment	Among consenting respondents	Welcome-page trigger upstream of consent	Confirm fielded timing and report difference
Four conditions	Control, generic, empathy, fiscal	Explicit route crosswalk in Part C	Use consistent substantive labels
Fair-fee input	Numeric current fee	Bracketed current-fee item	Describe proxy and covariate approximation
Fairness direction	Endpoints opposite to German item	-5 too low; +5 too high	Correct prose and audit directional results
Donation	Opt-outs zero; conditional analysis secondary	Original codes preserved in April data, lost in merged fields	Preserve explicit flags in final preparation
Exclusions	Consent; impossible values only	Non-test completed statuses; blanket sentinels	Document retention and variable-specific cleaning
Multiplicity	Holm within each outcome; FDR supplement	Holm appears in reanalysis code	Verify final tables contain both adjustments
Belief benchmark	True share shown by fiscal treatment	7.05% in code; 4.17% in GPX	Verify fielded values and recompute where needed
Linked moderation	Letter-by-survey interaction	Legacy and canonical linkage differ	Document canonical construction and rerun linked models

Topic	Local plan	Available implementation evidence	Required resolution
Exploratory analyses	Ideology, education, mechanism, optional moderators	Several existing appendix tables	Inventory completed, omitted, and infeasible checks

G.3 Coverage of planned analyses

The final package should account for each planned analysis even if it cannot be performed. This includes the Study 1 proportion comparison and logistic model; all three survey-versus-control contrasts on both primary outcomes; the empathy/fiscal versus generic contrasts; donation conditional on participation; lottery opt-out; invitation interactions; ideology and education moderation; the T2-versus-T3 mechanism comparison; income moderation; deprivation if linkable; prior-belief-gap analyses; baseline-fee and unfairness-direction moderation; possible control pooling; and descriptive manipulation checks.

A failed availability condition should be stated explicitly. For example, deprivation moderation can be marked unavailable if no approved area linkage exists. It should not quietly disappear from the plan comparison. Pooling T0 and T1, if undertaken, remains exploratory. A nonsignificant difference alone does not establish that the arms are equivalent.

H. Estimands and statistical methods

H.1 Invitation experiment

Let $L_i = 1$ denote an empathy invitation and $L_i = 0$ an efficiency invitation. For the status-defined participation outcome R_i, the primary descriptive effect is

$$\widehat{\Delta}_R = \overline{R}_{L=1} - \overline{R}_{L=0}.$$

Report this difference in percentage points, its two-sided 95% confidence interval, and the exact p-value. The existing analysis uses an uncorrected two-sample proportion test and a logistic regression of participation on invitation assignment. A logistic coefficient is a log odds ratio. Its exponentiation should be labelled an odds ratio, not a risk ratio or a percentage-point effect.

The existing code creates its point estimate as empathy minus efficiency. Its proportion-test object is called with efficiency first. The returned interval has the opposite contrast orientation. Before publication, negate and reverse the interval endpoints or compute the test with the intended group order. The manuscript's current point estimate and interval must be checked together.

Because message allocation is count-balanced within addressee target, a design-aligned sensitivity can account for that target. This would supplement the existing specification and should be labelled accordingly. Household administrative identifiers should not enter public output.

H.2 Survey experiment

Let T_{ik} indicate assignment to arm $k \in \{1, 2, 3\}$, with T0 as reference. The existing fair-fee model is

$$Y_i^{fee} = \alpha + \sum_{k=1}^{3} \beta_k T_{ik} + \gamma C_i + \varepsilon_i.$$

The existing primary donation model is

$$Y_i^{donation} = \alpha + \sum_{k=1}^{3} \beta_k T_{ik} + \varepsilon_i.$$

Each coefficient is an adjusted or unadjusted treatment-versus-control contrast in euros. Present outcome means, standard deviations, and observed Ns by arm separately from model Ns. For the fair-fee model, report the current-fee adjustment and its complete-case requirement. For donation, report the opt-out coding and the number of unresolved missing observations.

The reanalysis uses conventional model-based standard errors for its primary OLS output. Existing supplementary sensitivity tables also contain HC2 standard errors. Identify the method in each table. A confidence interval computed with robust standard errors should not be paired with a p-value from the conventional model without explanation.

The analysis is by assigned condition within the available outcome sample. Full-population intention-to-treat language requires an accounting of all initially assigned survey participants and missing outcomes. A completion-only analysis can be affected by treatment-related

attrition.

H.3 Contrasts and multiple comparisons

Within each primary outcome, the local plan places T1-T0, T2-T0, and T3-T0 in one Holm-adjusted family. This gives two families of three tests. Report raw and Holm-adjusted p-values. The plan also requests FDR-adjusted q-values across the six treatment-versus-control tests. Verify that the chosen FDR method is stated in the final code and table notes.

T2-T1 and T3-T1 are prespecified secondary comparisons without confirmatory family adjustment. T2-T3 is an exploratory mechanism comparison. The plan's wording calls some contrasts primary in one paragraph and secondary in another. The explicit multiplicity paragraph defines the implemented families. Disclose that interpretive choice and check it against the final registered record.

Confidence intervals should be labelled as pointwise unless simultaneous coverage was actually calculated. Holm-adjusting p-values does not automatically make ordinary 95% intervals simultaneous.

H.4 Interactions and interpretation

The invitation-interaction model adds L_i and $T_{ik}L_i$ terms. Its interaction coefficients describe differences in survey-treatment contrasts between invitation groups among analyzed respondents. Since entry into the survey follows invitation assignment, this is not automatically the joint population effect of the two stages.

Moderator models add the moderator and its interactions with all survey arms. Report the coding, scale, reference group, model N, interaction estimates, and intervals. Show marginal contrasts over meaningful moderator values if these are generated. A significant coefficient in one subgroup and a nonsignificant coefficient in another is not itself a test of their difference.

Ideology, education, income, and belief-gap analyses remain exploratory under the local plan. Inspect measurement timing. Sociodemographic questions occur after the experimental screens, so their reports should not be described without qualification as pre-treatment measurements, even when the underlying characteristics are stable.

H.5 Power, precision, and claims about weak effects

The local plan anticipated approximately 16,000 invitations, 10% take-up, and 1,600 survey respondents. It stated about 400 per arm and minimum detectable standardized effects near 0.20 for an unadjusted pairwise comparison and 0.23 under a conservative multiplicity threshold. These are planning statements from the local document. They are not a freshly verified power calculation.

The file named `power_sample_calculations.tex` concerns a later focus-group treatment

with pre/post measurement. It is not the power analysis for the letter experiment or four-arm Wave 1 experiment. It should not be attached as support for those designs.

[**AUTHOR ACTION H1**] Supply the original power calculation, assumptions, and stopping rule. If only the written plan survives, say so. A later design-sensitivity calculation should use the actual allocation and the outcome-specific model Ns, with all assumptions stated. It should be distinguished from a priori planning.

The final paper can state that an estimate is small or imprecise when supported by its interval. It should not infer zero effects from a p-value above .05. A formal equivalence analysis needs substantively justified bounds. A Bayes-factor analysis needs stated priors. Neither should be added solely to obtain a preferred conclusion.

I. Supplementary results: existing material and completion plan

I.1 Status of statistical tables

The current manuscript already includes seven generated supplementary tables. Their headers attribute generation to `analysis/results/results.qmd`, a path not found in the inspected workspace. They are preserved in place. This draft does not certify them as newly reproduced outputs or insert old estimates under changed sample definitions.

Existing table	Contents	Verification needed before final inclusion
A1	Stated-only fair fee, fee-bracket adjustment, HC2 results	Outcome coding, model Ns, robust inference
A2	Financing satisfaction, fairness, conditional donation	Scale direction, conditional selection, actual Ns
A3	Invitation-by-survey interactions	Canonical linkage construction and regenerated models
A4	Prior-belief gap	Verified fiscal benchmark and associative interpretation
A5	Ideology moderation	Coding, timing, standardization, model Ns
A6	Education moderation	Ordered-code assumption and other-category handling
A7	Income moderation	Missingness and ordered-code assumption

Once the source and instrument decisions are fixed, the final supplement should contain the following additional tables and figures. Their numbers are proposed and can be aligned with the manuscript at integration.

I.2 Required descriptive and design tables

Table I1: frame and allocation. Raw rows, blank identifiers, duplicate extra rows, retained invitation records, addressee targets, message arms, and any dispatched additional batches. Separate pre-assignment cleaning from post-assignment losses.

Table I2: participant flow. Initial survey assignments, consent, exposure, completion, usable outcome, and model inclusion by arm. Provide reasons where observed. Unknown cells remain explicit until the all-disposition export is available.

Table I3: characteristics by assigned arm. For each available characteristic, show measurement timing, mean/proportion, SD where meaningful, nonmissing N, and missingness. For Study 1, use administrative variables available for the whole randomized frame. Characteristics measured only among survey respondents cannot establish baseline comparability in the invitation population.

Table I4: outcome distributions. For each arm, show N, mean, SD, median, interquartile range, minimum, maximum, and the zero share where relevant. A fair-fee plot should display its skew and the concentration created by bracket representatives. A donation plot should show all eleven allocation categories and explicit opt-outs separately.

Table I5: missingness and retention. Report outcome and covariate missingness by arm. Distinguish routing, explicit refusal, valid zero, and unclassified missingness. Where initial assignments are available, examine retention by assigned treatment. Describe limitations of adjustment based on observed characteristics.

I.3 Confirmatory and planned secondary tables

Table I6: primary contrasts. For both primary outcomes, report all three arm-versus-control estimates, SEs, pointwise intervals, raw p-values, Holm-adjusted p-values, cross-outcome FDR q-values, and model N. State the current-fee adjustment for the fair-fee outcome.

Table I7: planned secondary contrasts. Report T2-T1 and T3-T1 for each primary outcome, with raw p-values and a secondary-analysis label. Show T2-T3 separately as exploratory.

Table I8: lottery decisions. Report explicit opt-out rates by arm and the planned logistic comparison. Show donation conditional on the participation proxy as a secondary analysis. Explain why it may reflect selection.

I.4 Robustness and exploratory analyses

Preserve the original outcome definition as the primary analysis. Supplement it with stated-only fair fees, alternative adjustment for current-fee brackets, and HC2 inference already represented in the manuscript materials. If a new bracket-bound sensitivity is run, state that it was added during revision. Do not silently recode the registered outcome.

For missing outcomes, prioritize a transparent account of the observed losses. Bounds or imputation can be useful only with explicit assumptions and an appropriate design. Treat hypothetical zeros, arbitrary imputations, and post-treatment compliance exclusions as different estimands rather than routine robustness.

Report the planned moderator analyses without selecting only favorable coefficients. If the data do not support deprivation linkage, explain its omission. If control pooling is performed, retain the four-arm confirmatory analysis and state the pooling rule. Include a complete inventory of attempted analyses in the replication README.

[**AUTHOR ACTION I1**] Locate the statistical-table generator, resolve the coding and linkage issues, rerun the final analysis from approved sources, and populate these sections. This document provides no new significance claims.

J. Consent, ethics, incentives, and disclosures

J.1 Ethical review status

Ethics applications and strategy documents are available locally. Their existence does not establish approval. Earlier descriptions sometimes refer to three survey conditions, whereas the stored instrument has four. The final account must identify the approved protocol, any amendments, and the relationship between the approved and implemented designs.

[**AUTHOR ACTION J1**] Insert the review body, decision date, reference number, approved version, and relevant amendments or exemption decision. Explain coverage of the invitation experiment, survey, lottery, and later recontact. Do not use the application date as the approval date.

J.2 Voluntary participation and consent

The welcome screen explains that participation is voluntary and that stopping or not participating brings no disadvantage for the respondent, child, or household. It gives an expected duration of 12-15 minutes. The data-use page requests agreement. A refusal triggers a confirmation question. Refusal at confirmation leads to the survey end under the stored route.

The survey separately asks permission to invite respondents to later activities and asks for contact details where permission is given. These later invitations must not be treated as required for inclusion in the present experiment. The lottery contact procedure is also distinct from consent to future research contact.

The invitation intervention precedes the online consent procedure. The final ethics account should explain the basis on which invitation framing and administrative participation tracking were authorized before respondents reached survey consent. Online consent to answer the survey does not by itself document that authorization.

J.3 Potential harms, deception, and debriefing

The final review should address possible discomfort from financial questions, emotional reactions to the narrative, and any concern that the municipal sender could affect willingness to participate. Describe the safeguards actually used. The welcome statement about no disadvantage is one documented safeguard.

The stored empathy debrief explicitly says that Sofia and her parents do not exist. The narrative is therefore fictional in the available instrument. The English draft frames its story as a newspaper excerpt. The stored German empathy screen does not use that introduction. The ethics account must follow the deployed material and explain the fictional presentation and its approval.

The GPX includes an empathy-only debrief route, filter 7602096 to page 7601898. It states: "Der Text über Sofia und Ihre Eltern ist typisch für viele Familien in METROPOLIS. Sofia und Ihre Eltern gibt es jedoch nicht." A working translation is: "The text about Sofia and her parents is typical of many families in METROPOLIS. However, Sofia and her parents do not exist." This establishes that a debrief was programmed. It does not establish that every assigned respondent reached it. [**AUTHOR ACTION J2**] Verify deployment, placement, exposure, and ethics approval of this debrief. Do not insert a blanket declaration of no

deception.

J.4 Compensation and delivery of incentives

The instrument advertises 25 prizes of EUR 100 for Wave 1. The maximum advertised prize pool is EUR 2,500. An individual participant's expected return depends on the number of eligible lottery participants, the draw rules, and the chosen allocation. The appendix cannot infer that expected return from the prize amount alone.

The later online discussion invitation advertises EUR 30 for an approximately one-hour discussion. That payment belongs to a later component. It is not guaranteed payment for completing this survey.

[**AUTHOR ACTION J3**] Supply the lottery eligibility rule, draw date and procedure, winner-contact procedure, number of prizes paid, aggregate donation transfer, and justification of the incentive arrangement. Keep identifying payment records outside the public replication package.

J.5 Confidentiality and data handling

The consent material promises confidentiality and separates answers from personal contact data. The project nevertheless uses linkable survey codes and later contact information. The technical description should distinguish pseudonymized working data from an anonymized public release. A unique code or administrative identifier is not rendered anonymous by removing a name column.

The final account should describe who held the municipal frame, who held the code key,

who could access contact details, how linkage was authorized, and when identifiers were separated or deleted. Report the actual approved retention and access arrangements. Do not infer compliance from a boilerplate GDPR statement.

Public materials should exclude family identifiers, survey access codes, names, addresses, telephone numbers, email addresses, and identifying free text. Review indirect identifiers and rare combinations before release. Where original raw records cannot be made public, agree a documented access solution with the journal rather than making an unsupported promise of unrestricted availability.

J.6 Funding, roles, interests, and AI assistance

[**AUTHOR ACTION J4**] Insert the funding source and grant identifier, each funder's role, the municipality's role in frame provision and mailing, and any publication-review arrangements. Complete the journal's author-specific interest disclosures. No absence of conflicts is assumed in this draft.

Suggested AI-assistance disclosure for author verification: "The authors used OpenAI Codex on 16 October 2026 to assist with locating journal guidance, organizing study documentation, drafting supplementary text, and preparing working English translations of German study materials. [Insert the verified model/version and access information.] The authors reviewed and corrected the final text and translations." Add any other actual uses. Do not claim that review is complete until it is.

J.7 APSA principles statement: completion template

"The study was reviewed by [body, reference, date]. The approved protocol covered [components]. Participation was voluntary. The invitation intervention and administrative participation tracking were authorized through [basis]. The survey requested consent through [procedure]. Participants received [actual incentive arrangement and justification]. The narrative [verified status and disclosure]. Debriefing [actual process or justification]. Data were protected through [actual separation, access, retention, and release arrangements]. Potential harms and the municipal recruitment relationship were addressed by [actual safeguards]."

This is a completion template. It is not yet an affirmation of compliance. The resolved statement should address respect for persons, beneficence, and justice in concrete terms.

K. Replication materials and source provenance

K.1 Proposed deposit

The final deposit should distinguish restricted original data from public analytical inputs and public materials. Include a README, a data dictionary, the original instruments, a documented preparation workflow, ordered analysis scripts, output logs, and the final tables and figures. State software versions, package versions, the random seeds where relevant, and the commands needed to reproduce the article and appendix.

The public workflow must not rerun a live mailing randomization or overwrite source workbooks. Supply an appropriately protected demonstration of allocation if necessary. The archived mailing assignment is the experimental record. Regenerating a new assignment is

not a substitute for it.

The data dictionary must distinguish original response codes from cleaned values. In particular, preserve the distinction between explicit lottery opt-out and missing response, exact amounts and bracket proxies, original assignment numbers and substantive arm names, and canonical versus legacy linkage fields.

K.2 Data availability statement: completion template

"The data, code, and materials required to reproduce the analyses in this article and its supplementary appendix are available in the Journal of Experimental Political Science Dataverse at [DOI]. [State any restricted files, reasons, and access procedure.]"

Do not insert a DOI until a deposit exists. The final article should place its Data Availability Statement before the bibliography, as requested in the journal's submission instructions.

K.3 Local provenance register

The following register identifies the local evidence used for this draft. Paths are relative to the project root. Checksums are SHA-256 hashes calculated on 16 October 2026. They establish which bytes were inspected. They do not establish approval, registration, or completeness of a file.

Full hashes and relative paths are in `source_manifest.sha256`. The shortened values below are the first 16 hexadecimal characters. The manifest contains no individual records.

Source number	Evidence	SHA-256 prefix
1	Manuscript snapshot	ebbb2b7d96c39f29
2	Existing design and count audit	f4e4767d1bef2aee
3	Local preregistration comparison	515426cfb0f8f655
4	Dated registration draft	6b093b5fc7851044
5	Existing preparation and analysis	bf025517504162a8
6	Canonical allocation algorithm	0d82f2ebd08fd9bb
7	Private administrative frame	a9b5c38852e13344
8	Private final assignment slice	3cbeba58c93aa654
9	Private final assignment slice	5d48b2a0d696cb06
10	Original survey export	99296f4c36c6aa02
11	Original status export	3ed66ffe87be579a
12	Instrument and assets	731112b4998617ab
13	Question wording and categories	2caa65fe8e0ea2e9
14	Merged snapshot	f12c8a1c6230f271
15	Final invitation paragraphs	e0a0170c57d2e27b
16	Alternative English stimulus draft	b64c2b88ed769667
17	Consent draft	021cc5d15a837568
18	Ethics application, not approval	3967e1e2faa26c8b

The entries follow the same order as the full relative paths in `source_manifest.sha256`. Keep that manifest with this author draft. Private data files are not distributed with the appendix.

L. Reporting-checklist response matrix

This matrix follows the components of the journal's official reporting checklist. Row labels are abbreviated. Answers refer to the available evidence and the sections above. An open item is not marked complete merely because a section heading exists.

L.1 Hypotheses

Component	Response and location
Research questions	A.1-A.3 describe invitation participation, fair-fee preferences, and donation.
Specific expectations	A.3 and G distinguish the primary expectations, cumulative invitation interaction, and exploratory moderators. Verify final registered wording.

L.2 Participants and context

Component	Response and location
Eligibility and rationale	B.1 describes the municipal frame. A.1 explains the substantive population. Exact administrative eligibility criteria remain open.
Exclusions	B.1 distinguishes pre-assignment cleaning. E.5 and F document retained completed-status records and outcome-specific missingness.
Recruitment changes	B.3 requests dispatch and additional-batch records. Changes during recruitment have not been established.
Recruitment process	B.2-B.3 describe addressee selection, letters, codes, and online access. Final printed common material is outstanding.
Survey provider	A.1 identifies Unipark/EFS as the platform. No commercial panel recruitment is documented for this study. The city supplied the mailing frame.
Field dates and follow-up	B.3 requests actual dispatch, opening, closing, and reminder dates. Later activities are outside this paper's outcome scope. File dates are not substituted for field dates.
Setting	A.1 and C.3: METROPOLIS family frame, municipal mailing, online questionnaire, municipal and other kindergarten providers.

Component	Response and location
Participation denominator	B.4 gives counts and calculation. Delivered-mail and all-disposition denominators remain open.

L.3 Assignment

Component	Response and location
Sequence generation	B.2 specifies the canonical script, seed, operation order, and random shuffling. C.2 describes the survey trigger.
Allocation restrictions	B.2 documents balancing within addressee target. C.2 records survey `balanced=0`.
Assignment unit	A.2: retained family identifier for Study 1 and participant record for Study 2. No group-assigned survey treatment is documented.
Evidence from implementation	B.2 and C.1 provide allocation totals and route mapping. Deployment history and initial survey assignments remain to be verified.
Covariate comparison	I.2 specifies the table to generate. It is not supplied as if already calculated. Measurements collected only from respondents cannot establish Study 1 population balance.
Unequal block probabilities	Addressee assignment depends on availability of ZPF2. Message assignment is balanced within target. No unequal-probability survey blocks are established.
Blinding	C.4: no tested blinding process established. Visibility of the assigned message and standardized delivery are disclosed.

L.4 Intervention delivery

Component	Response and location
All conditions	C.1 and D provide control and all three screen packages, including images.
Replicable instructions	D reproduces invitation paragraphs and German screen text. M supplies static text and complete codebook wording. Common printed letter and browser screenshots are outstanding.
Delivery channel and timing	B.3 and C.2-C.3 distinguish postal invitation, online assignment, consent, stimulus, and outcomes.
Software	A.1 identifies Unipark/EFS and the stored 24.1 requirement. K requests the reproducible analysis environment and versions.
Repetition and order	A.2 and C.3 describe two sequential components. No practice-round or within-person sequence of all treatment screens is documented.
Other embedded activities	C.3 and J.4 disclose the later recontact invitation before the lottery. Focus groups and Wave 2 are outside the current paper.
Repeat participation	The source retains unique survey IDs and canonical invitation codes. Browser re-entry and cross-session assignment persistence still need deployment verification.
Duration and sessions	J.2 records advertised survey duration. Actual duration distributions have not been audited. Later discussion duration is kept separate.
Session counts	This is an asynchronous survey, not a laboratory session design. No laboratory group-rotation counts apply. Initial survey assignments remain unknown.
Fictional presentation	J.3 documents the fictional family and programmed empathy debrief. Approval and actual debrief exposure remain to be verified.
Delivery anomalies	D.5, E.4-E.5, F, and G document stimulus-version, fiscal-benchmark, coding, timing, and linkage discrepancies.

Component	Response and location
Instruction quizzes	C.4 and M document a post-screen recognition item. No separate instruction quiz is established.
Practice tasks	No practice task is documented for these two components.
Debriefing and understanding	J.3 supplies the programmed empathy debrief. C.4 distinguishes recognition from verified understanding.
Research-team observation	No observation of individual screen exposure or reading is established. Request page logs in C.4.
Manipulation checks	C.4 and E.1 identify `v_57`. Descriptive arm-level results remain to be generated.
Incentive administration	E.3 and J.4 describe the advertised lottery and donation. Actual drawing and transfers remain open.

L.5 Measures and analysis

Component	Response and location
Primary and secondary definitions	E provides variables, coding, euro conversion, and fair-fee formula.
Composite construction	E.2 defines the fair-fee proxy and E.3 defines donation. No new multi-item solidarity index is introduced.
Exact questionnaire	M transcribes the complete codebook and static screens. D supplies treatment text and translations.
Planned versus later analyses	A.3, G, H.3, and I state labels and an analysis inventory. Final registration verification remains open.
Eligibility and pre-assignment loss	B.1 reports raw frame size, blank IDs, repeated rows, and final frame.
Initial allocation	B.2 supplies letter counts. C.2 and F.4 explain why retained survey-arm counts are not initial assignments.
Receipt of intervention	C.4 and F.4 request exposure logs and nonreceipt reasons. No assumption of complete receipt is made.

Component	Response and location
Attrition and missing outcomes	F provides retained and usable sample counts. Initial survey attrition and reasons cannot be calculated from completion-only records.
Analysis inclusion	E.5 and F.1 give outcome and covariate requirements, including the 32 additional fair-fee covariate losses.
Arm summaries	B.4 and F give available counts. Means, SDs, and distributions are specified for regeneration in I.2.
Analysis level and uncertainty	H identifies family-record and respondent-level analyses, contrast scales, conventional SEs, and HC2 sensitivities.
Attrition predictors	I.2 specifies analysis after recovering the initial-assignment population. No completed attrition analysis is claimed.
Covariate missingness	E.5 and F document current issues. I.2 requests full arm-level missingness and model-deletion counts.
Missing-data method	Existing models use complete cases. Explicit lottery opt-outs are a prespecified zero coding, not generic missing-value imputation.
Subgroup deletions	I requires model-specific N for every interaction and moderator. Counts have not been inferred from total survey N.
Weighting	No survey weighting is present in the inspected primary model specifications. No claim of city-population representativeness follows.

L.6 Other reporting

Component	Response and location
Ethics decision	J.1: application present; final decision details not established.
Registration access	G.1 and N: local text available; immutable registered record and timestamp outstanding.
Funding and role	J.6 requests grant details, funder and municipal roles, restrictions, and publication-review arrangements.
Competing interests	J.6 leaves author-specific disclosures open.
Replication access	K proposes the deposit and data statement. No public DOI is claimed.

M. Full questionnaire and static instrument text

M.1 Scope and transcription conventions

The transcription below covers the full local Wave 1 codebook dated 16 April 2026. It contains questions and response categories, not respondent answers. Page identifiers and variable names are retained for replication. Blank options and platform labels are reproduced rather than silently repaired. Two embedded JavaScript fragments are omitted from the readable transcription. They remain in the original codebook. Their code is not respondent-facing question wording. The presence of blank options in the export does not establish whether they were visible or selectable in the browser.

The accompanying static-screen transcription supplies explanatory text omitted from the codebook, including welcome, consent, and lottery instructions. Plain-text extraction does not reproduce browser layout. Routing details are documented separately from visible text.

Final screenshots and the complete sanitized platform export should accompany the replication materials.

Static explanatory screens and stored filters

Container depot (page 7566844)

Standard component: No depot component is assigned to this reference question yet.

W1 (page 7599681)

Leben mit Kindergarten-Kindern in METROPOLIS

Wir freuen uns, dass Sie die Befragung gestartet haben!

Ihre Erfahrung als Eltern in METROPOLIS zählt. Mit Ihrer Teilnahme helfen Sie uns, die aktuelle Situation von Familien mit Kindergarten-Kindern in Ihrer Stadt besser zu verstehen. Die Befragung dauert etwa 12-15 Minuten. Als Dankeschön können Sie einen von 25 Gutscheinen im Wert von jeweils 100€ gewinnen.

Diese Befragung ist Teil einer rein wissenschaftlichen Untersuchung, mit der wir das Leben bei Ihnen vor Ort verbessern wollen. Wir verfolgen keinerlei gewinnorientierte Ziele.

Ihre Teilnahme ist freiwillig. Sie können zu jeder Zeit die Teilnahme abbrechen. Wenn Sie nicht teilnehmen, hat das keinerlei Nachteile für Sie, Ihr Kind oder Ihren Haushalt.

Falls Sie das Fenster versehentlich schließen: Rufen Sie die Website erneut auf und geben Sie Ihren Code ein, um die Befragung fortzusetzen.

Details zur Verlosung finden Sie am Ende des Fragebogens, darunter auch die Möglichkeit, für einen guten Zweck zu spenden.

Diese Befragung ist Teil einer größeren Studie. Wir laden Sie am Ende dieser Befragung herzlich ein, weiter daran teilzunehmen.

Für das Forschungsteam der [UNIVERSITY]

Prof. Dr. [RESEARCHER NAME]

Bitte klicken Sie auf „Weiter“, um zu beginnen.

W2 (page 7599683)

Datenschutz und Kontakt

Alle Ihre Antworten und Prozessdaten (z.B. Ihre Antwortzeiten) werden streng vertraulich und in Übereinstimmung mit der Datenschutz-Grundverordnung (DS-GVO) behandelt.

Wir Forscher kennen Ihren Namen oder Ihre Adresse nicht und werden diese auch niemals kennen. Die Stadt METROPOLIS hat für uns den Brief an Sie versandt und Ihnen eine Codenummer gegeben. Wir wissen anhand dieser Codenummer nur, dass Sie in METROPOLIS wohnen und (mindestens) ein Kindergarten-Kind haben.

Ihre Antworten in dieser Befragung werden anonymisiert. Das bedeutet, dass

alle Antworten getrennt von Ihren personenbezogenen Daten gespeichert und verarbeitet werden. Es können keine Rückschlüsse auf Ihre Person gezogen werden. Nur die anonymisierten Daten werden für die Auswertung genutzt.

Wenn Sie Fragen zu diesem Fragebogen oder zu der „Leben mit Kindergarten-Kindern in METROPOLIS“-Befragung haben, kontaktieren Sie uns bitte über die unten angegebenen Kontaktdaten.

Herr [RESEARCHER NAME] Tel: [PHONE NUMBER] E-Mail: [CONTACT EMAIL]

W2_Conf (page 7601324)

Datenschutz und Kontakt

Alle Ihre Antworten und Prozessdaten (z.B. Ihre Antwortzeiten) werden streng vertraulich und in Übereinstimmung mit der Datenschutz-Grundverordnung (DS-GVO) behandelt.

Wir Forscher kennen nicht Ihren Namen oder Ihre Adresse und werden diese niemals kennen. Die Stadt METROPOLIS hat für uns den Brief an Sie versandt und Ihnen eine Codenummer gegeben. Wir wissen anhand dieser Codenummer nur, dass Sie (mindestens) ein Kind haben, das in einem Kindergarten in METROPOLIS betreut wird.

Ihre Antworten in dieser Befragung werden anonymisiert. Das bedeutet, dass alle Antworten getrennt von Ihren personenbezogenen Daten gespeichert und ve-

rarbeitet werden. Es können keine Rückschlüsse auf Ihre Person gezogen werden. Nur die anonymisierten Daten werden für die Auswertung genutzt.

Wenn Sie Fragen zu diesem Fragebogen oder zu der „Leben mit Kindergarten-Kindern in METROPOLIS“-Befragung haben, kontaktieren Sie uns bitte über die unten angegebenen Kontaktdaten.

Herr [RESEARCHER NAME] Tel: [PHONE NUMBER] E-Mail: [CONTACT EMAIL]

Final page (page 7601337)

Die Befragung ist hiermit beendet.

Wir danken Ihnen dennoch für Ihr Interesse an unserer Studie.

Im Namen des ganzen Forschungsteams Ihr

Prof. Dr. [RESEARCHER NAME] von der [UNIVERSITY]

W3 (page 7600425)

Beginnen wir mit einigen Fragen zu Ihrem Leben mit Kindergarten-Kind oder Kindergarten-Kindern in METROPOLIS.

lttr_ignr (page 7601803)

Es kann sein, dass Sie noch einen weiteren Brief mit Bitte um Teilnahme an der Studie bekommen haben oder bekommen werden.

Diesen brauchen Sie nicht weiter zu beachten.

kitafee (page 7601862)

Denken Sie bitte wieder an Ihr Kind, das als letztes Geburtstag hatte.

cost_brdenprnt (page 7601860)

In jedem Monat entstehen für Kindergärten Kosten für die Gebäude und Freiflächen, Ausstattung, Unterhalt und Personal.

fairness1 (page 7601863)

Denken Sie bitte wieder an Ihr Kindergarten-Kind, das als letztes Geburtstag hatte.

Sie haben angegeben, dass Sie für dieses Kind Elternbeiträge in Höhe von monatlich #v_180# zahlen.

sys_spprt (page 7601865)

In METROPOLIS stellt die Stadt Kita-Plätze zur Verfügung. Diese werden finanziert durch…

Steuergelder aus ganz NRW,

Mittel aus dem Haushalt der Stadt,

Elternbeiträge.

Sociodemographics (page 7601064)

Nun folgen einige Fragen zu Ihrer Person.

cntryclt (page 7601879)

Sie haben angegeben, dass Sie internationale Wurzeln haben.

consentdtlnk (page 7601892)

Für uns ist es wichtig, Ihre Antworten mit Informationen der Stadt METROPOLIS zur Finanzierung Ihres Kindergartenplatzes zusammenzubringen.

Das dürfen wir nur nach Ihrer Erlaubnis mithilfe einer Codenummer. Mit dieser können wir die Daten verknüpfen, ohne je Ihren Namen oder Ihre Adresse zu kennen. Nach der Verknüpfung löschen wir die Codenummer für immer. Auch bei Ihrer Zustimmung können keinerlei Informationen auf Sie persönlich zurückgeführt werden.

Ihre Zustimmung ist freiwillig und kann jederzeit zurückgenommen werden.

consentscnd (page 7601893)

Wir sind nun am Ende der Befragung angekommen.

Vielen Dank für Ihre Teilnahme!

Bevor wir zu den Details der Verlosung kommen: Für die Aussagekraft unserer Studie ist Ihre weitere Unterstützung extrem wertvoll.

Wir möchten Sie zu zwei zukünftigen Bestandteilen der Studie einladen:

1. In wenigen Wochen organisieren wir mit einigen Teilnehmenden dieser Umfrage Online-Videogespräche von etwa 60 Minuten Länge. In kleiner Runde (etwa sechs Eltern) möchten wir tiefergehend über Ihre Erfahrungen als Kindergarteneltern sprechen. Die Teilnahme an diesen Video-Gesprächen aus dem eigenen Zuhause wird mit 30 Euro vergütet.

2. In einigen Monaten kommt eine zweite Befragung von etwa 10 Minuten Länge. Für die Teilnahme werden Sie die Möglichkeit erhalten, erneut bei einer Verlosung für 25 mal 100 Euro teilzunehmen.

donation (page 7601895)

Als Dankeschön für Ihre Mühen nehmen Sie an der Verlosung von 25 mal 100 Euro teil.

Wenn Sie 100 Euro gewinnen, können Sie Ihren Gewinn zum Teil oder ganz spenden. Diese Spende geht an den Verein [ORGANISATION] in METROPOLIS, der damit Projekte in Kindergärten in METROPOLIS finanziert. Der

Verein arbeitet in METROPOLIS, versorgt bedürftige Kinder mit gesunder Kost, stellt Kindern in Armut anregendes Fördermaterial wie Bücher zur Verfügung und schafft eine kinderfreundliche Umwelt (weitere Informationen: [ORGANISATION WEBSITE]).

Den Betrag, den Sie behalten, erhalten Sie als digitalen Geschenkgutschein.

Falls ich gewinne, behalte ich: 100 €.

debrief (page 7601898)

Der Text über Sofia und Ihre Eltern ist typisch für viele Familien in METROPOLIS.

Sofia und Ihre Eltern gibt es jedoch nicht.

Final page (page 7566845)

Wir bedanken uns ganz herzlich für Ihre heutige Teilnahme an unserer Befragung.

Im Namen des ganzen Forschungsteams

Ihr

Prof. Dr. [RESEARCHER NAME] von der [UNIVERSITY]

Wenn Sie noch Fragen haben, kontaktieren Sie uns bitte gerne: Herr [RESEARCHER NAME] Tel: [PHONE NUMBER] E-Mail: [CONTACT EMAIL]

Stored filter conditions

These are platform condition codes, retained as recorded. Code 3 corresponds to equality in the treatment routes and code 6 to inequality. Other operator codes are left uninterpreted here. `fvkn` and bracket flags determine combinations. The complete platform export is needed to reproduce complex routing.

datenschutz (7601332). variable v_1, operator 3, value 2, connector 0, brackets 0/0.

data_conf (7601335). variable v_126, operator 3, value 2, connector 0, brackets 0/0.

Empathy Treatment Filter (7601924). variable c_0001, operator 3, value 2, connector 0, brackets 0/0.

Community of Solidarity Treatment Filter (7601925). variable c_0001, operator 3, value 3, connector 0, brackets 0/0.

Thematic Prime Treatment Filter (7601926). variable c_0001, operator 3, value 4, connector 0, brackets 0/0.

Mnpltnchck Filter (7601927). variable c_0001, operator 6, value 1, connector 0, brackets 0/0.

Fairness2 Filter (7601930). variable v_65, operator 6, value 6, connector 0, brackets 1/0; variable v_65, operator 4, value 11, connector 1, brackets 0/1.

sys_spprt_reason Filter (7608694). variable v_68, operator 4, value 11, connector 0, brackets 0/0.

Has Partner? (7602970). variable v_85, operator 3, value 1, connector 0, brackets 0/0.

Cntryclt Filter (7601931). variable v_99, operator 4, value 3, connector 0, brackets 0/0.

Weitere Studieneinladung Filter (7601934). variable v_108, operator 3, value 1, connector 0, brackets 0/0.

contactdonate, contactdonate_eml, contactdonate_mob Filter (7601971). variable v_168, operator 6, value 88, connector 0, brackets 0/0.

Empathy Debrief Filter (7602096). variable c_0001, operator 3, value 2, connector 0, brackets 0/0.

M.2 Full codebook transcription

WP5 Field Experiment - Invitation Survey

Project ID | 1092096

Survey URL | [SURVEY URL]

Date | 2026-04-16 09:00:29

1 Page: System (PGID 7566843)

User-defined variable (- Type 931)

c_0001 | c_0001 | bigint | treatment_id

c_0002 | c_0002 | varchar | Code2

3 Page: W1 (PGID 7599681)

4 Page: W2 (PGID 7599683)

Stimmen Sie dieser Nutzung Ihrer Antwortdaten zu? (q_14098121 - Type 111)

v_1 | v_1 | int | W2Q

1 | Ja

2 | Nein

5 Page: datenschutz (PGID 7601332)

5.1 Page: W2_Conf (PGID 7601324)

Ohne Ihre Zustimmung dürfen wir Ihre Antworten nicht auswerten. Die Befragung würde hier enden. Bitte bestätigen Sie Ihre Entscheidung. Stimmen Sie dieser Nutzung Ihrer Antwortdaten zu? (q_14101541 - Type 111)

v_126 | v_126 | int | conf

1 | Ja

2 | Nein

5.2 Page: data_conf (PGID 7601335)

6 Page: W3 (PGID 7600425)

7 Page: chld_resp (PGID 7601800)

Wer trifft die Entscheidungen rund um den Kindergartenbesuch Ihres Kindes/Ihrer Kinder (z. B. Auswahl der Einrichtung, der Betreuungszeiten)? (q_14100327 - Type 111)

v_24 | v_24 | int | chld_resp

1 | Hauptsächlich oder ausschließlich ich

2 | Ich zusammen mit jemand anderem

3 | Hauptsächlich oder ausschließlich jemand anderes

4

88 | Möchte ich nicht beantworten

8 Page: chld_rltn (PGID 7601801)

Jetzt zu Ihrem Kind, das aktuell einen Kindergarten in METROPOLIS besucht oder bald besuchen wird. Falls Sie mehrere Kindergarten-Kinder haben, denken Sie bitte an das Kind, das als letztes Geburtstag hatte. Was ist Ihre Beziehung zu diesem Kind, das als letztes Geburtstag hatte? (q_14100330 - Type 111)

v_26 | v_26 | int | chld_rltn

1 | Leibliche Mutter

2 | Leiblicher Vater

3 | Stiefmutter

4 | Stiefvater

6 | Andere Beziehung, und zwar:

7

88 | Möchte ich nicht beantworten

v_27 | v_27 | varchar | Andere Beziehung, und zwar:

9 Page: chld_othr (PGID 7601802)

Wie viele Kinder leben in Ihrem Haushalt, die im Kindergartenjahr 2026-27 (also spätestens ab dem 01. August 2026) einen Kindergarten in METROPOLIS besuchen werden? (q_14100336 - Type 111)

v_29 | v_29 | int | chld_othr

1 | %s Kind(er)

2

88 | Möchte ich nicht beantworten

v_30 | v_30 | varchar (with type check: Integer) | %s Kind(er)

10 Page: lttr_ignr (PGID 7601803)

11 Page: lifesat (PGID 7601804)

Alles in allem, wie zufrieden sind Sie mit Ihrem Leben in Ihrer Stadt oder Ihrem Dorf? (q_14100347 - Type 111)

v_32 | v_32 | int | lifesat

1 | 1 Äußerst unzufrieden

2 | 2

3 | 3

4 | 4

5 | 5

6 | 6

7 | 7

8 | 8

9 | 9

10 | 10

11 | 11 Äußerst zufrieden

12

88 | Möchte ich nicht beantworten

99 | Weiß nicht

12 Page: polint (PGID 7601805)

Wie stark interessieren Sie sich für Politik in METROPOLIS? (q_14100990 - Type 111)

v_34 | v_34 | int | polint

1 | Überhaupt nicht interessiert

2 | Wenig interessiert

3 | Ziemlich interessiert

4 | Sehr interessiert

6

88 | Möchte ich nicht beantworten

13 Page: leftright (PGID 7601806)

In der Politik spricht man manchmal von “links” und “rechts”. Wo auf dieser Skala würden Sie sich selbst einstufen, wenn 1 für links steht und 11 für rechts? (q_14100996 - Type 111)

v_37 | v_37 | int | leftright

1 | 1 Links

2 | 2

3 | 3

4 | 4

5 | 5

6 | 6

7 | 7

8 | 8

9 | 9

10 | 10

11 | 11 Rechts

12

88 | Möchte ich nicht beantworten

99 | Weiß nicht

14 Page: myrsat (PGID 7601807)

Alles in allem, wie zufrieden sind Sie mit dem Kindergarten, den Ihr Kind besucht? (q_14101001 - Type 111)

v_39 | v_39 | int | myrsat

1 | 1 Äußerst unzufrieden

2 | 2

3 | 3

4 | 4

5 | 5

6 | 6

7 | 7

8 | 8

9 | 9

10 | 10

11 | 11 Äußerst zufrieden

12

88 | Möchte ich nicht beantworten

99 | Weiß nicht

15 Page: kitafee (PGID 7601862)

Wie hoch ist der monatliche Elternbeitrag, den Sie zurzeit für die Betreuung dieses Kindes im Kindergarten zahlen? (q_14110630 - Type 111)

v_180 | v_180 | int | kitafee

1 | 0 €

2 | 1 - 50 €

3 | 51 - 100 €

4 | 101 - 150 €

5 | 151 - 200 €

6 | 201 - 250 €

7 | 251 - 300 €

8 | 301 - 350 €

9 | 351 - 400 €

10 | 401 - 450 €

11 | 451 - 500 €

12

88 | Möchte ich nicht beantworten

99 | Weiß nicht und kann ich nicht schätzen

16 Page: polsolchld_duisnow, polsolchld_nrwnow, polsolchld_duisfut, polsolchld_nrwfut (PGID 7601051)

Für einen Kindergartenplatz in METROPOLIS zahlen Eltern wie Sie aktuell zwischen 0 und etwa 500 Euro.Denken Sie nun an andere Kindergarten-Kinder neben Ihrem/-n eigenen. Wie sehr stimmen Sie den folgenden Aussagen zu oder nicht zu? Ich bin bereit, die Gesamtkosten von Kindergartenplätzen mitzutragen für … (q_14101006 - Type 311)

v_48 | v_48 | int | … alle anderen Kindergarten-Kinder in METROPOLIS, während mein Kind oder meine Kinder noch im Kindergarten sind.

1 | Stimme überhaupt nicht zu

2 | Stimme eher nicht zu

3 | Teils/teils

4 | Stimme eher zu

5 | Stimme voll und ganz zu

v_49 | v_49 | int | … alle anderen Kindergarten-Kinder im restlichen NRW, während mein Kind oder meine Kinder noch im Kindergarten sind.

1 | Stimme überhaupt nicht zu

2 | Stimme eher nicht zu

3 | Teils/teils

4 | Stimme eher zu

5 | Stimme voll und ganz zu

v_50 | v_50 | int | … alle anderen Kindergarten-Kinder in METROPOLIS, wenn mein Kind oder meine Kinder nicht mehr im Kindergarten sind.

1 | Stimme überhaupt nicht zu

2 | Stimme eher nicht zu

3 | Teils/teils

4 | Stimme eher zu

5 | Stimme voll und ganz zu

v_51 | v_51 | int | … alle anderen Kindergarten-Kinder im restlichen NRW, wenn mein Kind oder meine Kinder nicht mehr im Kindergarten sind.

1 | Stimme überhaupt nicht zu

2 | Stimme eher nicht zu

3 | Teils/teils

4 | Stimme eher zu

5 | Stimme voll und ganz zu

(q_14106539 - Type 111)

v_167 | v_167 | int | No response polsol

88 | Möchte ich nicht beantworten

17 Page: cost_brdenprnt (PGID 7601860)

Bitte schätzen Sie: Wie groß ist der Anteil der Gesamtkosten der Kindergärten in METROPOLIS, der durch die Eltern der Kindergarten-Kinder jeden Monat getragen wird?

(q_14101009 - Type 111)

v_53 | v_53 | int | cost_brdenprnt

1 | Eltern tragen die Kosten zu %s %.

2

88 | Möchte ich nicht beantworten

99 | Weiß nicht und kann ich nicht schätzen

v_56 | v_56 | varchar (with type check: Integer) | Eltern tragen die Kosten zu %s %.

18 Page: Empathy Treatment Filter (PGID 7601924)

18.1 Page: Empathy Treatment (PGID 7571736)

19 Page: Community of Solidarity Treatment Filter (PGID 7601925)

19.1 Page: Community of Solidarity Treatment (PGID 7576345)

20 Page: Thematic Prime Treatment Filter (PGID 7601926)

20.1 Page: Thematic Prime Treatment (PGID 7571738)

21 Page: Mnpltnchck Filter (PGID 7601927)

21.1 Page: mnpltnchck (PGID 7601861)

Worum ging es in dem Text hauptsächlich? (q_14101012 - Type 111)

v_57 | v_57 | int | mnpltnchck

1 | Um Fakten über Kindergärten in METROPOLIS.

2 | Um die Erfahrungen einer Familie aus METROPOLIS mit der Kindergartenbetreuung.

3 | Um die Rolle der Steuerzahler bei der Finanzierung von Kindergartenplätzen.

4 | Um den Tagesablauf in Kindergärten in METROPOLIS.

5 | Um die für die Jahreszeiten typischen Aktivitäten in Kindergärten in METROPOLIS.

22 Page: fairness1 (PGID 7601863)

Ist Ihr Elternbeitrag für den Kindergarten dieses Kindes gerecht oder ist er Ihrer Meinung nach ungerechterweise zu niedrig bzw. ungerechterweise zu hoch? (q_14101024 - Type 111)

v_65 | v_65 | int | fairness1

1 | -5 Ungerechterweise zu niedrig

2 | -4

3 | -3

4 | -2

5 | -1

6 | 0 Gerecht

7 | +1

8 | +2

9 | +3

10 | +4

11 | +5 Ungerechterweise zu hoch

12

88 | Möchte ich nicht beantworten

23 Page: Fairness2 Filter (PGID 7601930)

23.1 Page: fairness2 (PGID 7601864)

Sie haben angegeben, dass Sie den Betrag eher ungerecht empfinden. Was wäre Ihrer Meinung nach ein gerechter Monatsbeitrag? (q_14101025 - Type 111)

v_66 | v_66 | int | fairness2

1 | %s € monatlich.

2

88 | Möchte ich nicht beantworten

v_67 | v_67 | varchar (with type check: Integer) | %s € monatlich.

24 Page: sys_spprt (PGID 7601865)

Wie zufrieden sind Sie mit der Art, wie die Stadt METROPOLIS Kindergartenplätze finanziert? (q_14101026 - Type 111)

v_68 | v_68 | int | sys_spprt

1 | 1 Äußerst unzufrieden

2 | 2

3 | 3

4 | 4

5 | 5

6 | 6

7 | 7

8 | 8

9 | 9

10 | 10

11 | 11 Äußerst zufrieden

12

88 | Möchte ich nicht beantworten

99 | Weiß nicht

25 Page: sys_spprt_reason Filter (PGID 7608694)

25.1 Page: sys_spprt_reason (PGID 7605106)

Bitte nennen Sie uns den wichtigsten Grund für diese Bewertung: (q_14110295 - Type 142)

v_178 | v_178 | blob | sys_spprt_reason_open

(q_14110633 - Type 121)

v_182 | v_182 | int | Möchte ich nicht beantworten

0 | not quoted

1 | quoted

26 Page: Sociodemographics (PGID 7601064)

27 Page: gndr (PGID 7601867)

Sie sind … (q_14101033 - Type 111)

v_71 | v_71 | int | gndr

1 | männlich

2 | weiblich

3 | divers

4

88 | Möchte ich nicht beantworten

28 Page: brthyr (PGID 7601868)

In welchem Jahr sind Sie geboren (z. B. 1987)? (q_14101035 - Type 111)

v_73 | v_73 | int | brthyr

1 | Geben Sie bitte Ihr Geburtsjahr an:

2

88 | Möchte ich nicht beantworten

v_74 | v_74 | varchar (with type check: Integer) | Geben Sie bitte Ihr Geburtsjahr an:

29 Page: schedu (PGID 7601869)

Welchen höchsten allgemeinbildenden Schulabschluss haben Sie? Wenn Sie ihren höchsten Schulabschluss im Ausland gemacht haben, versuchen Sie sich bitte den vorgegebenen Kategorien zuzuordnen. (q_14101037 - Type 111)

v_76 | v_76 | int | schedu

1 | Schüler/in an einer allgemeinbildenden Schule

2 | von der Schule abgegangen ohne Schulabschluss

3 | Hauptschulabschluss, Volksschulabschluss oder vergleichbarer Abschluss

4 | Polytechnische Oberschule der DDR mit Abschluss der 8. oder 9. Klasse

5 | Realschulabschluss, Mittlere Reife oder vergleichbarer Abschluss

6 | Polytechnische Oberschule der DDR mit Abschluss der 10. Klasse

7 | Fachhochschulreife

8 | Abitur, Allgemeine oder Fachgebundene Hochschulreife, Gymnasium bzw. EOS, auch EOS mit Lehre

9 | einen anderen Schulabschluss und zwar

10

88 | Möchte ich nicht beantworten

v_77 | v_77 | varchar | einen anderen Schulabschluss und zwar

30 Page: dsbrg_years (PGID 7601870)

Wie viele Jahres Ihres Lebens haben Sie insgesamt bisher in METROPOLIS gewohnt? (q_14101038 - Type 111)

v_79 | v_79 | int | dsbrg_years

1 | %s Jahre

2

88 | Möchte ich nicht beantworten

99 | Weiß nicht

v_80 | v_80 | varchar (with type check: Integer) | %s Jahre

31 Page: postcode (PGID 7601871)

Wie lautet die Postleitzahl Ihres aktuellen Wohnortes (z. B. 47051)? // (q_14101040 - Type 111)

v_82 | v_82 | int | postcode

1

2

88 | Möchte ich nicht beantworten

99 | Weiß nicht

v_83 | v_83 | varchar (with type check: Integer)

32 Page: partner (PGID 7601872)

Haben Sie im Moment einen Partner oder eine Partnerin? (q_14101042 - Type 111)

v_85 | v_85 | int | partner

1 | Ja

2 | Nein

3

88 | Möchte ich nicht beantworten

33 Page: Has Partner? (PGID 7602970)

33.1 Page: livtogr (PGID 7601873)

Lebt Ihr Partner oder Ihre Partnerin mit Ihnen im selben Haushalt? (q_14101043 - Type 111)

v_87 | v_87 | int | livtogr

1 | Ja

2 | Nein

3

88 | Möchte ich nicht beantworten

34 Page: hhsize (PGID 7601874)

Wie viele Personen leben ständig in Ihrem Haushalt, Sie selbst eingeschlossen? (q_14101044 - Type 111)

v_89 | v_89 | int | hhsize

1 | %s Personen

2

888 | Möchte ich nicht beantworten

v_90 | v_90 | varchar (with type check: Integer) | %s Personen

35 Page: hhchldu18 (PGID 7601875)

Und wie viele der Personen in Ihrem Haushalt sind Kinder oder Jugendliche, das heißt Personen im Alter von unter 18 Jahren? (q_14101045 - Type 111)

v_92 | v_92 | int | hhchldu18

1 | %s Personen

2

888 | Möchte ich nicht beantworten

v_93 | v_93 | varchar (with type check: Integer) | %s Personen

36 Page: hhincome (PGID 7601876)

Wie hoch ist das monatliche Gesamteinkommen Ihres Haushalts aus allen Quellen nach Steuern und Abzügen? (q_14101046 - Type 111)

v_95 | v_95 | int | hhincome

1 | 0€ - 1500€

2 | 1501€ - 2500€

3 | 2501€ - 3500€

4 | 3501€ - 4500€

5 | 4501€ oder mehr

11

88 | Möchte ich nicht beantworten

99 | Weiß nicht

37 Page: emplymnt (PGID 7601877)

Was ist Ihr aktueller Beschäftigungsstatus? (q_14101047 - Type 111)

v_97 | v_97 | int | emplymnt

1 | In bezahlter Arbeit (angestellt, selbstständig, im Familienunternehmen tätig)

2 | In Elternzeit

3 | Hausmann/ Hausfrau, Pflege von Kindern oder Angehörigen

4 | In Ausbildung/ Studium/ Weiterqualifizierung

5 | Arbeitslos/ Arbeitssuchend

6 | In Rente/ Pension/ Arbeitsunfähig

7 | Etwas anderes

8

88 | Möchte ich nicht beantworten

38 Page: mgrtnstts (PGID 7601878)

Haben Sie internationale Wurzeln? (q_14101048 - Type 111)

v_99 | v_99 | int | mgrtnstts

1 | Ja, ich selbst bin nach Deutschland eingewandert, und zwar erstmals mit %s Jahren.

2 | Ja, ich bin in Deutschland geboren und mindestens ein Elternteil ist im Ausland geboren.

3 | Ja, meine Eltern und ich sind in Deutschland geboren und mindestens ein Großelternteil ist im Ausland geboren.

4 | Nein, ich habe keine internationalen Wurzeln.

5

88 | Möchte ich nicht beantworten

99 | Weiß nicht

v_100 | v_100 | varchar (with type check: Integer) | Ja, ich selbst bin nach Deutschland

eingewandert, und zwar erstmals mit %s Jahren.

39 Page: Cntryclt Filter (PGID 7601931)

39.1 Page: cntryclt (PGID 7601879)

Gibt es ein oder mehrere Herkunftsländer neben Deutschland, das oder die Sie in Ihrem Leben stark geprägt hat oder haben? (q_14101050 - Type 111)

v_102 | v_102 | int | cntryclt

1 | Ja, und zwar:

2 | Nein, es gibt kein Herkunftsland, das mich stark geprägt hat.

3

88 | Möchte ich nicht beantworten

v_103 | v_103 | varchar | Ja, und zwar:

40 Page: consentdtlnk (PGID 7601892)

Stimmen Sie der Verknüpfung Ihrer Antworten mit den Informationen der Stadt zur Finanzierung Ihres Kindergartenplatzes zu? (q_14101053 - Type 111)

v_107 | v_107 | int | consentdtlnk

1 | Ja, ich stimme der Verknüpfung zu

2 | Nein, ich stimme nicht zu

41 Page: consentscnd (PGID 7601893)

Dürfen wir Sie zu diesen Bestandteilen der Studie einladen? (q_14101054 - Type 111)

v_108 | v_108 | int | consentscnd

1 | Ja, ich erlaube Ihnen, mich wieder einzuladen.

2 | Nein, laden Sie mich nicht wieder ein.

42 Page: Weitere Studieneinladung Filter (PGID 7601934)

42.1 Page: contactscnd_eml, contactscnd_mob (PGID 7601894)

Danke für Ihre Erlaubnis! Wir würden Sie gerne per SMS und per E-Mail einladen. Unter welcher Mobilfunknummer und E-Mailadresse möchten Sie eingeladen werden? (q_14102849 - Type 121)

v_135 | v_135 | int | Laden Sie mich unter dieser E-Mail-Adresse ein:

0 | not quoted

1 | quoted

v_136 | v_136 | varchar (with type check: E-mail address) | Laden Sie mich unter dieser E-Mail-Adresse ein:

v_137 | v_137 | int | Laden Sie mich unter dieser Mobilfunknummer ein:

0 | not quoted

1 | quoted

v_138 | v_138 | varchar (with type check: Phone number) | Laden Sie mich unter dieser Mobilfunknummer ein:

v_141 | v_141 | int | Ich möchte nicht wieder eingeladen werden.

0 | not quoted

1 | quoted

43 Page: donation (PGID 7601895)

Wie möchten Sie im Gewinnfall Ihre 100 Euro aufteilen?Falls ich gewinne, spende ich: (q_14106913 - Type 111)

v_168 | v_168 | int | donation_amount_single_choice

1 | 0 €

2 | 10 €

3 | 20 €

4 | 30 €

5 | 40 €

6 | 50 €

7 | 60 €

8 | 70 €

9 | 80 €

10 | 90 €

11 | 100 €

12

88 | Ich möchte nicht an der Verlosung teilnehmen.

44 Page: contactdonate, contactdonate_eml, contactdonate_mob Filter (PGID 7601971)

44.1 Page: contactdonate, contactdonate_eml, contactdonate_mob (PGID 7601896)

Falls Sie einen Gutschein gewinnen, benötigen wir eine Kontaktmöglichkeit.Wie möchten Sie im Gewinnfall benachrichtigt werden? (q_14101057 - Type 111)

v_116 | v_116 | int | contactdonate, contactdonate_eml, contactdonate_mob

1 | Kontaktieren Sie mich unter der Nummer oder der E-Mail-Adresse, die ich Ihnen gerade gegeben habe.

2 | Nutzen Sie folgende E-Mail-Adresse:

3 | Nutzen Sie folgende Mobilfunknummer:

4

88 | Ich möchte nicht an der Verlosung teilnehmen.

v_117 | v_117 | varchar (with type check: E-mail address) | Nutzen Sie folgende E-Mail-Adresse:

v_118 | v_118 | varchar (with type check: Phone number) | Nutzen Sie folgende Mobilfunknummer:

45 Page: contact_info (PGID 7601897)

Möchten Sie per E-Mail über die zentralen Ergebnisse dieses Projekts informiert werden? (q_14103240 - Type 111)

v_155 | v_155 | int | question_text

(q_14103226 - Type 111)

v_154 | v_154 | int | vorherige_mail

1 | Ja, an die bereits angegebene E-Mail-Adresse.

(q_14101058 - Type 111)

v_120 | v_120 | int | contact_info

2 | Ja, an folgende E-Mail-Adresse:

3 | Nein, informieren Sie mich nicht.

v_121 | v_121 | varchar (with type check: E-mail address) | Ja, an folgende E-Mail-Adresse:

46 Page: Empathy Debrief Filter (PGID 7602096)

46.1 Page: debrief (PGID 7601898)

47 Page: FinalOpenQuestion (PGID 7608693)

Gibt es noch irgendetwas, was Sie uns mitteilen wollen? (q_14116705 - Type 142)

v_187 | v_187 | blob | FinalComments

N. Local preregistration comparison text

The following is the full local `field_exp_code/prereg.md` text used in preparing the comparison. It is reproduced as a source document. Statements about collecting no data are claims in that source, not independently verified statements made by this appendix. Replace or supplement this transcription with the immutable registered version and its timestamp before submission.

> Preregistration Template from AsPredicted.org Data collection No, no data have been collected for this study yet.
>
> Hypothesis This project has two linked randomized components implemented in one field rollout. Study 1 (invitation letter take-up) Does a prosocial, empathy-framed invitation letter increase survey take-up (accessing the survey link and providing consent) relative to a government efficiency invitation? H1 (take-up): The empathy-framed letter increases take-up compared to the placebo letter. Study 2 (within-survey information treatments) Among those who enter the Wave 1 survey, do short informational materials about municipal kindergarten financing affect (a) fairness preferences about parental fees, (b) support for col-

lective financing, and (c) a behavioral solidarity measure (donation decision)? We test whether solidarity-relevant information increases willingness to accept higher fees as fair, increases support for taxpayer financing, and increases voluntary donations. • T0 Control: No additional information screen. • T1 Generic information about METROPOLIS kindergartens/fees (neutral, descriptive). • T2 Empathy story vignette about a child benefiting from the solidarity system. • T3 Fiscal information about taxpayer contributions and the gap between total costs and parental payments.

H1 (information treatments): Relative to T0, the solidarity-relevant or empathy treatments T2 and T3 increase perceived fairness of higher parental fees and increase willingness to donate. We also report the pre-specified secondary contrasts T2 vs T1 and T3 vs T1. H2 (cumulative empathy): Treatment effects of T2/T3 are stronger among respondents who received the empathy-framed invitation letter. H3 (political orientation, exploratory): Left–right self-placement moderates information effects. We expect stronger positive effects of T2/T3 (on fair fee and donation) among more left-leaning respondents than among more right-leaning respondents (treatment × ideology interaction). H4 (education, exploratory): Education moderates information effects. We expect treatment effects of T2/T3 to differ by education level (treatment × education): higher-educated respondents respond more (less) strongly to the informational community of solidarity treatment (emotional empathy treatment) than lower educated parents.

Dependent variable Study 1 primary dependent variable (take-up) • Take-up

indicator: 1 if the invited household accesses the survey and provides informed consent to participate in Wave 1; 0 otherwise (tracked via unique participant codes). Study 2 primary outcomes (measured in Wave 1 survey) Primary attitudinal outcome: Fair fee preference We use a single confirmatory attitudinal dependent variable: "Fair monthly fee (EUR)". Construction of "Fair monthly fee (EUR)" (pre-specified rule) Participants answer three items: 1. Current monthly parental fee (numeric). 2. fairness1: Perceived fairness of current fee on -5 (unfairly too high) to +5 (unfairly too low), 0 = fair. 3. fairness2: If fairness1 $\neq$ 0, respondent states what would be a fair monthly fee (numeric). We define "Fair monthly fee (EUR)" as follows: • If fairness1 = 0, then Fair monthly fee = current monthly fee. • If fairness1 $\neq$ 0, then Fair monthly fee = fairness2. Because current fees in METROPOLIS follow an income-based sliding scale, the fair monthly fee variable is mechanically correlated with income for respondents where fairness1 = 0 (their current fee stands as the outcome value). We therefore include current monthly fee as a covariate in all Study 2 regressions involving this outcome. Primary behavioral outcome: Solidarity donation Donation amount from hypothetical lottery winnings: 0–100 EUR in steps of 10. Exploratory attitudinal outcomes: - Support for the current financing arrangement Single item: satisfaction with how METROPOLIS finances kindergarten places (1–11 scale).

Conditions Study 1 (population-level letter randomization; 2 arms; 50/50) • Letter A: goverment efficiency invitation text. • Letter B: Empathy/prosocial

community invitation text. Study 2 (within-survey information randomization; 4 arms; equal allocation) • T0 Control: No additional information screen. • T1 Generic information about METROPOLIS kindergartens/fees (neutral, descriptive). • T2 Empathy story vignette about a child benefiting from the solidarity system. • T3 Fiscal information about taxpayer contributions and the gap between total costs and parental payments. Randomization is individual-level within the survey among respondents who provide consent to Wave 1 participation.

Analyses General approach: • Primary analyses are intention-to-treat (participants analyzed by assigned condition). • We report effect sizes with 95% confidence intervals and two-sided tests. Study 1 (take-up) • Difference in proportions between empathy letter and placebo letter. • Logistic regression: Take Up ~ Empathy letter. Study 2 (within-survey outcomes) Primary estimand: average treatment effects relative to control (T0). • Continuous outcomes: regression Outcome ~ T(1,2,3; T0 is reference), plus arm means and mean differences. • Donation amount among lottery participants: All Wave 1 respondents, with lottery opt-outs coded as 0 EUR. Secondary analysis: conditional on lottery participation. Lottery opt-out is treated as a secondary outcome because it may itself be affected by treatment. • Lottery opt-out: logistic regression Lottery opt out ~ T1 + T2 + T3. Planned contrasts: • Primary contrasts: T2 vs T0 and T3 vs T0; T2 vs T1 and T3 vs T1. • Benchmark contrast: T1 vs T0. • We will still report all three treatment vs control comparisons.

Outliers and Exclusions Study 1: No observations will be excluded. All randomized households will be included in the take-up analysis. Study 2: only respondents who provide informed consent will be included in survey analyses. No additional exclusions will be made based on treatment compliance, response patterns, or outcome values. No outcome-based outlier removal will be applied. For numeric variables, only clearly impossible values outside pre-specified feasible ranges (for example, negative fee amounts) will be treated as missing.

Sample Size • Study 1: All eligible households receiving the municipal invitation letter (approximately 16,000 households) are included in randomization and take-up measurement. • Study 2: All respondents who access the survey and provide consent during the field period are included and randomized. For Study 2, expected N depends on take-up rates; with ~16,000 letters and expected take-up of 10%, we anticipate approximately 1,600 respondents (~400 per arm), yielding a minimum detectable effect of d $\approx$ 0.20 for unadjusted pairwise comparisons and d $\approx$ 0.23 under Holm correction (worst case, $\alpha/3$), at $\alpha = 0.05$ with 80% power. • Data collection ends when the pre-defined field period closes (based on the municipality's mailing schedule and survey closing date).

Other H3 and H4 from Section 2 are treated as exploratory. We will test treatment $\times$ ideology and treatment $\times$ education interactions using OLS with interaction terms, but these analyses are not subject to confirmatory correction. H5 (mechanism, exploratory): We expect T2 (empathy vignette) to show relatively stronger effects on the behavioral donation outcome compared to T3, while T3

(fiscal information) shows relatively stronger effects on the attitudinal fair-fee outcome compared to T2. We test this by comparing T2 vs T3 effect sizes separately on each primary outcome; no correction is applied. Exploratory heterogeneity analyses

Income: If an income item is collected with sufficient response rate, we will explore whether treatment effects differ by income. These analyses are exploratory due to expected item non-response and measurement noise. Area deprivation: If an area-level deprivation indicator is available and linkable, we will explore whether treatment effects differ between more and less deprived areas. Prior-belief gap: Before treatment, respondents estimate the parent vs taxpayer financing split (%). We define a belief gap as the absolute difference between this estimate and the true split shown in the information arm. This moderator is only applicable within T3 (the only arm that reveals the true financing split). Within T3, we expect larger effects on fair monthly fee and donation among respondents with a larger belief gap (tested via belief-gap interactions within T3 only). Other planned secondary checks • Baseline fee level and perceived unfairness direction (fairness1 < 0 vs > 0) as additional exploratory moderators. • Placebo check / pooling of controls (exploratory): Compare T0 and T1 on the two primary outcomes. If statistically and substantively indistinguishable, we may pool T0 and T1 for exploratory robustness analyses; confirmatory analyses follow the original four-arm assignment. • Manipulation checks will be used descriptively (no causal mediation claims).

Data handling and exclusions • Study 1: Analyze take-up for all randomized households. • Study 2: Include only respondents who provide Wave 1 consent. Multiple comparisons We pre-specify one primary outcome for Study 1: • Take-up For Study 2, we pre-specify two primary outcomes: • (A) Fair monthly fee (EUR) • (B) Donation amount (EUR) For each Study 2 primary outcome separately, the confirmatory family consists of the three treatment-versus-control tests: • T1 vs T0 • T2 vs T0 • T3 vs T0 Within each outcome, these three p-values will be adjusted using the Holm procedure. We will report both raw p-values and Holm-adjusted p-values. As a transparent supplement, we will also report false discovery rate adjusted q-values across all confirmatory Study 2 primary-outcome tests. The pre-specified secondary contrasts T2 vs T1 and T3 vs T1 will be reported without confirmatory multiple-testing correction and interpreted as secondary analyses. Documentation • Final versions of letter texts and the four information screens will be archived (PDF/screenshot) at the time of preregistration, before rollout. • Randomization procedures and allocation ratios will be documented (including survey-platform randomizer settings export where possible).

Power statement • The projected minimum detectable effects for Study 2 are expressed in standardized units for power-calculation purposes only. Substantive results will be reported primarily as treatment effects in original outcome units.

Name [PROJECT NAME]: Information, solidarity, and payment behavior in municipal kindergarten fees

Type of Project Experiment

Other No data